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25th DNV NORDIC MARITIME UNIVERSITIES WORKSHOP

Technical University of Denmark (DTU), Kgs. Lyngby

30-31 January 2025

Program and Book of Abstracts



FOTO THOMAS HJORT JENSEN

Venue: DTU's Meeting Center

Anker Engelunds Vej 1, 2800 Kgs. Lyngby

Preface

The DNV Nordic Maritime Universities Workshop is organized as a collaboration between DNV and universities in the Nordic region with a maritime related education or research line. The workshop covers all research topics related to naval architecture, maritime engineering and maritime transport, including safety, energy efficiency and environmental performance, environmental pressures, new technologies and digitalization.

The 25th Nordic Maritime Universities Workshop was held on 30-31 January 2025 at the Technical University of Denmark (DTU), Lyngby Campus. The workshop has been organized and hosted by the Maritime Group at the Department of Civil and Mechanical Engineering (DTU Construct). In total we received 77 abstracts from 7 countries. This includes 23 abstracts from Denmark, 23 from Sweden, 16 from Norway, 10 from Germany, 3 from Finland, 1 from The Netherlands, and 1 from Poland. The presentation of the abstracts and the talks is carried out over two days of the workshop and in 10 sessions, distributed over 7 topics:

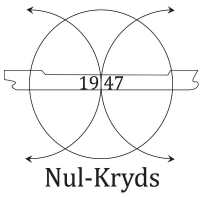
- Maritime Safety & Risk Reduction (17 talks)
- Structures & Ship Design (8 talks)
- Numerical Methods & Marine Hydrodynamics (14 talks)
- Ship Operations & Navigation (14 talks)
- Autonomous Shipping & Digitalization (8 talks)
- Alternative Marine Fuels (8 talks)
- Wind Assisted & Alternative Propulsion (8 talks)

This year a special issue has been initiated in *International Shipbuilding Progress* to commemorate the 25th Nordic Maritime Universities Workshop. All abstract presenters have been invited to submit a full paper, to be considered for publication in this journal after a peer-review process. This compendium includes the workshop program, the session details and the 77 abstracts arranged in alphabetical order.

Mostafa Amini-Afshar , host and chair of the workshop at DTU.
Erik Vanem, responsible for the workshop at DNV.

Acknowledgment

The Maritime Group at DTU Construct is very grateful to the *Danish Society for Naval Architecture and Marine Engineering (DSNAME - Skibsteknisk Selskabs Fond)*, for their generous financial support of this workshop. In addition, we appreciate the help we received from the student maritime club at DTU, *Nul-Kryds*, in organizing this workshop.





Program Overview

Thursday, 30th of January

- 08:00-08:30: **Registration** - (M1).
- 08:30-08:35: **Welcome & Intro.**. Mostafa Amini-Afshar & Erik Vanem - (M1).
- 08:35-08:45: **DNV Company presentation**, Tomas Tronstad - (M1).
- 08:45-09:15: **Keynote 1:** *Arising Challenges and Opportunities for Assuring Maritime Software Reliability in the AI Era*. By Dr. Shuai Wang, DNV - (M1).
- 09:15-09:45: **Keynote 2:** *On the Design of Bridges Against Ship Collisions*. By Professor Emeritus Preben Terndrup Pedersen, DTU - (M1).
- 09:45-10:00: **Coffee break**
- 10:00-12:00: **Session 1**, Maritime Safety & Risk Reduction I - (S1).
- 10:00-12:00: **Session 2**, Structures & Ship Design - (M1).
- 10:00-12:00: **Session 3**, Numerical Methods & Marine Hydrodynamics I - (S9).
- 12:00-13:00: **Lunch** - (Faculty Club).
- 13:00-15:00: **Session 4**, Ship Operations & Navigation I - (M1).
- 13:00-15:00: **Session 5**, Autonomous Shipping & Digitalization - (S1).
- 15:00-15:30: **Coffee break**
- 15:30-17:00: **Session 6**, Ship Operations & Navigation II - (M1).
- 15:30-17:00: **Session 7**, Numerical Methods & Marine Hydrodynamics II - (S1).
- 17:00-17:30: Steering Committee Meeting - (M1).
- 19:00: **Dinner, Sponsored by DNV** - (Peter Lieps Hus)

Friday, 31st of January

- 08:00-08:30: **Tea & Coffee** - (M1).
- 08:30-09:00: **Keynote 1:** *Maritime Forecast* by Jens Christian Petersen DNV - (M1).
- 09:00-09:30: **Keynote 2:** *Marine Transport of CO_2 in Solid Form.* By Dr. Henrik O. Madsen, Co-founder and CEO - DecarbonICE - (M1).
- 09:30-10:00: **Coffee break**
- 10:00-12:00: **Session 8**, Maritime Safety & Risk Reduction II - (S1).
- 10:00-12:00: **Session 9**, Alternative Marine Fuels - (M1).
- 12:00-13:00: **Lunch** - (Faculty Club)
- 13:00-15:00: **Session 10**, Wind Assisted & Alternative Propulsion - (M1).
- 15:00-15:15: **Closure of the Workshop** - (M1).
  - Special issue invitation: Peter Wellens, Delft University of Technology.
  - Next NMU Workshop: To the Paris of the North - University of Tromsø (UiT) The Arctic University of Norway.
  - Closing Remarks: Mostafa Amini-Afshar & Erik Vanem.

Session Details

Session 1, (10:00-12:00) - Thursday (S1)
Maritime Safety & Risk Reduction I
Chair: Associate Professor Marie Lützen

T.1, (10:00-10:10). Olga Kazimierska, Gdańsk University of Technology ..... Page 30.
A novel method for maritime risk assessment based on first principles.

T.2, (10:10-10:20). Swantje Krupp, (NTNU) ..... Page 38.
Coastal green energy supply stations.

T.3, (10:20-10:30). Chamila Ranasinghe, KTH Royal Inst. of Tech. ..... Page 53.
Development of a mission-suitability assessment framework for marine vehicles (M-Score).

T.4, (10:30-10:45). Gabriel Chikelu, Aalto University ..... Page 15.
Developing a Framework to Assess and Enhance Port Preparedness for Extreme Weather Events.

T.5, (10:45-11:00). Jørgen Dalen, Oslo School of Architecture and Design ..... Page 16.
Feedback on results vs feedback on behaviour. What works best for reducing consumption?

T.6, (11:00-11:15). Pahansen de Alwis, Research Institutes of Sweden ..... Page 17.
Cross-Disciplinary Insights for Maritime Safety: Applying Epidemiology and Biomechanics in Marine Craft Design and Operation.

T.7, (11:15-11:30). Karl Garme, KTH Royal Institute of Technology ..... Page 24.
Experimental investigation of biomechanical response to shock and vibration exposure at high speed at sea.

T.8, (11:30-11:45). Ogechukwu Nwafor, KTH Royal Institute of Technology ..... Page 48.
On GPU-acceleration of non-linear ship motion simulations in time-domain.

T.9, (11:45-12:00). Michele Acciaro, Copenhagen Business School ..... Page 1.
Exploring Regenerative Business Models for Biodiversity Restoration in the Shipping Industry.

Session 2, (10:00-12:00) - Thursday (M1)
Structures & Ship Design
Chair: Professor Jonas Ringsberg

T.1, (10:00-10:15). Chao Ye, KTH Royal Institute of Technology ..... Page 72.
Integrated Shipbuilding Framework for Decarbonisation Based on Digital Twins

T.2, (10:15-10:30). Oweis Al-Karawi, Chalmers University of Technology ..... Page 4.
Structural Integrity and Crashworthiness of Ships Powered by Small Modular Reactors.

T.3, (10:30-10:45). Hadi Amlashi, University of South-Eastern Norway ..... Page 5.
Durability and Performance of Offshore Wind Turbines under Fatigue Conditions.

T.4, (10:45-11:00). Marten Beiler, German Aerospace Center ..... Page 10.
Development of an inline inspection process for automated fatigue strength assessment of welded steel joints with data driven methods.

T.5, (11:00-11:15). Xiao Lang, Chalmers University of Technology ..... Page 40.
Physics-guided metamodel for vertical bending-induced fatigue damage monitoring in container vessels.

T.6, (11:15-11:30). Bonaventura Tagliafierro, Uppsala University ..... Page 57.
A Discrete Element Method-based Mooring Dynamic Solver for Complex Soil–Cable Interaction.

T.7, (11:30-11:40). Jan Luca Störkel, German Aerospace Center ..... Page 56.
Integration of the h-AES method into the design process of large structures for the fracture mechanical evaluation of accidental loads.

T.8, (11:40-11:50). Mahamudul Hasan Tanvir, German Aerospace Center ..... Page 59.
Investigating Material Behavior and Fatigue Strength of Welded Joints at Cryogenic Temperatures.

Session 3, (10:00-12:00) - Thursday (S9)

Numerical Methods & Marine Hydrodynamics I

Chair: Professor Rickard Bensow

T.1, (10:00-10:15). Minas Argyros, FORCE Technology ..... Page 7.
Systematic Analysis of the Effects of Scale and Draft Variation on Form Factor Using a Combined EFD/CFD Method.

T.2, (10:15-10:30). Peter Wellens, Delft University of Technology ..... Page 67.
Further steps towards generalisation of green water probability distributions on ships with forward speed.

T.3, (10:30-10:45). Qais Shehadeh Khraisat, Chalmers Uni. of Technology ..... Page 31.
Assessment of Blockage Effect on Cavitation Dynamics and Underwater Radiated Noise: A Numerical Study.

T.4, (10:45-11:00). Rui Lopes, Chalmers University of Technology ..... Page 41.
Influence of turbulence model selection on model-scale propeller and rudder performance.

T.5, (11:00-11:15). Joseph Praful Tomy, MAN Energy Solutions ..... Page 61.
Numerical and Experimental Investigation of Noise-generating Mechanisms for a Full-Scale Propeller.

T.6, (11:15-11:30). Yunfeng Ding, (DTU) ..... Page 19.
Numerical Modelling of 3D gap resonances between side-by-side barges under combined wave-current excitation.

T.7, (11:30-11:45). Gergely Kovacs, (DTU) ..... Page 37.
CFD and experimental study of a model-scale two-body hinged raft wave energy converter.

T.8, (11:45-12:00). Zhilong Wei, (DTU) ..... Page 66.
An Experimental Study on the Hydrodynamic Drag of Flexible Seaweed Mimics in a Uniform Current.

Session 4, (13:00-15:00) - Thursday (M1)
Ship Operations & Navigation I
Chair: Associate Professor Claes Eskilsson

T.1, (13:00-13:15). Rasmus Benjamin Ruff, (DTU) ..... Page 54.
Separating the impact of Hull and Propeller Degradation on Ship Resistance with Thrust Measurements.

T.2, (13:15-13:30). Jaike van Twiller, IT University of Copenhagen ..... Page 63.
Deep Reinforcement Learning for Revenue Management under Uncertainty in Master Stowage Planning on Container Vessels.

T.3, (13:30-13:45). Hui Xue, Arctic University of Norway ..... Page 70.
Impact of attention and working memory on maritime navigational performance and situational awareness.

T.4, (13:45-14:00). Mingyang Zhang, Aalto University ..... Page 76.
AI-Driven Ship Brain Framework for Intelligent Decision Making in Sustainable Maritime Navigation.

T.5, (14:00-14:15). Chengqian Zhang, Chalmers University of Technology ..... Page 74.
Hydrologic Riverbed Generation and Operational Analysis for Autonomous Vessels in Meandering Inland Waterways.

T.6, (14:15-14:30). Martin Baerveldt, (NTNU) ..... Page 8.
Tracking Vessels Using Multiple Extended Object Tracking.

T.7, (14:30-14:45). Dhaneswara Al Amien, Nord University ..... Page 2.
Modeling autonomus shipping diffusion: an agent-based modelling.

T.8, (14:45-15:00). Hoang Anh Tran, (NTNU) ..... Page 62.
Parallel distributed collision avoidance with intention consensus based on ADMM.

Session 5, (13:00-15:00) - Thursday (S1)
Autonomous Shipping & Digitalization
Chair: Associate Professor Ulrik D. Nielsen

T.1, (13:00-13:15). Alberto Tamburini, (DTU) ..... Page 58.
Reduce fleet emissions through a bi-objective tramp ship routing and scheduling problem.

T.2, (13:15-13:30). Casper A. Drangsfeldt, Uni. of Southern Denmark ..... Page 21.
Data Utilization for In-situ Condition-Based Maintenance of Working Vessels in Offshore Wind.

T.3, (13:30-13:45). Faraz Eftekhar, Ultrabulk A/S ..... Page 22.
Use of Machine Learning for Estimation of Wave Added Resistance and its Application in Ship Performance Analysis.

T.4, (13:45-14:00). Chiara Giovannini, KTH Royal Institute of Technology ..... Page 26.
Machine Learning for Ship Hull Design.

T.5, (14:00-14:15). Tae-Eun Kim, The Arctic University of Norway ..... Page 32.
Can self-driving car rules streamline autonomous ship regulations?

T.6, (14:15-14:30). Yaqub A. Prabowo, (DTU) ..... Page 51.
Quantifying Good Seamanship: A Framework for Risk-Aware Trajectory Planning in Autonomous Vessel Navigation.

T.7, (14:30-14:45). Raphaël E. G. Mounet, (DTU) ..... Page 44.
Sea State Estimation from Wave-Induced Ship Responses using Machine Learning.

T.8, (14:45-15:00). Mohammad Sheikholeslami, Chalmers Uni. of Tech ..... Page 55.
Free surface wave problems via physics-informed neural networks (PINNs).

Session 6, (15:30-17:00) - Thursday (M1)

Ship Operations & Navigation II

Chair: Associate Professor Karl Garme

T.1, (15:30-15:45). Daniel Vergara, Chalmers University of Technology ..... Page 64.
Short Sea Shipping Route Segmentation and Bayesian Optimisation of Engine Power Allocation.

T.2, (15:45-16:00). Rickard Bensow, Chalmers University of Technology ..... Page 11.
Research gaps towards achieving good environmental status in our oceans with respect to underwater noise.

T.3, (16:00-16:15). Sahil Bhagat, The Arctic University of Norway ..... Page 12.
Weighted measurement of ship navigation competence during loss of ship control.

T.4, (16:15-16:30). Sahil Mathur, Oslo School of Architecture and Design ..... Page 43.
Evaluating Iconography Design Guidelines for Navigation Control Function Icons: Systemic Design Approach to Enhancing Usability in Integrated Navigation Systems.

T.5, (16:30-16:45). Sheng Xu, The Arctic University of Norway ..... Page 71.
How can we enhance maritime simulation training using insights from comprehensive accident data?

T.6, (16:45-16:55). Elma Ramic, University of Southern Denmark ..... Page 52.
Addressing Seasickness of Technicians on board Working Vessels.

Session 7, (15:30-17:00) - Thursday (S1)

Numerical Methods & Marine Hydrodynamics II

Chair: Professor Harry B. Bingham

T.1, (15:30-15:45). Max Bitsch, (DTU) ..... Page 13.
A Second-Order WENO-based Finite Volume Method for Flooding Simulations using Shallow Water Equations with Enhanced Bathymetry Resolution.

T.2, (15:45-16:00). Jens Visbech, (DTU) ..... Page 65.
A Digital Twin Framework for Massively Parallel Simulations of Water Waves interacting with Large-Scale Offshore Structures.

T.3, (16:00-16:15). Claes Eskilsson, Aalborg University ..... Page 23.
Hybrid WNPF-LSTM Modelling of Wave-Body Problems.

T.4, (16:15-16:30). Mathias Klahn, Niels Bohr Institute (KU) ..... Page 36.
Heavy tails and high-order probability density functions for the surface elevation in irregular seas.

T.5, (16:30-16:40). Laurits T. Nicolaisen, (DTU) ..... Page 45.
Non-intrusive Reduced Order Modelling for Free Surface Flows.

T.6, (16:40-16:50). Gabor Albrect, Aalborg University ..... Page 3.
Wind-Wave Coupling for Improved Modelling of Coastal Flooding.

Session 8, (10:00-12:00) - Friday (S1)
Maritime Safety & Risk Reduction II
Chair: Associate Professor Peter Wellens

T.1, (10:00-10:15). Christopher Dirzka, Copenhagen Business School ..... Page 20.
Navigating uncertainty: The role of size and tie strength in shaping resilience for multinational corporations.

T.2, (10:15-10:30). Johannes Beiker, The German Aerospace Center ..... Page 9.
Goal/Risk-based Design of Methanol-powered Ships.

T.3, (10:30-10:45). Adanna N. Okonkwo, Aalto University ..... Page 47.
Improving Energy Savings Onboard Passenger Ship.

T.4, (10:45-11:00). Tae-eun Kim, The Arctic University of Norway ..... Page 33.
2nd-year results from the i-MASTER Horizon Europe Project.

T.5, (11:00-11:15). Benjamin Lagemann, (NTNU). ..... Page 39.
Research at NTNU towards net zero shipping.

T.6, (11:15-11:30). Jorgen Depken, German Aerospace Center ..... Page 18.
On the safety comparison of bunkering LH_2 and LNG.

T.7, (11:30-11:45). Filipe Landu Nzongo, Oslo Sch. of Arch. and Design ..... Page 46.
Steering new habits: How feedback can increase awareness of our energy usage. An integrated Systematic Review of Eco-feedback

T.8, (11:45-11:55). Eskil Kjemperud, University of South-Eastern Norway ..... Page 35.
What risks are covered with current safety practises in the Maritime business?

Session 9, (10:00-12:00) - Friday (M1)
Alternative Marine Fuels
Chair: Associate Professor Yanlin Shao

T.1, (10:00-10:15). Chi Zhang, Chalmers University of Technology ..... Page 75.
Identification of tank cleaning operations at sea based on machine learning method using AIS data.

T.2, (10:15-10:30). Seyed Parsa Parvasi, (DTU) ..... Page 49.
Ports as Energy Hubs: Renewable Energy Integration and Distribution.

T.3, (10:30-10:45). Fayas Malik Kanchiralla, Chalmers Uni. of Technology ..... Page 28.
Shipping fleet energy transition: Well to wake assessment model.

T.4, (10:45-11:00). Shaghayegh Kazemi Esfeh, German Aerospace Center ..... Page 29.
Liquid Hydrogen as a potential decarbonization solution for a large cruise ship.

T.5, (11:00-11:15). Tyge Kjær, Roskilde University ..... Page 34.
The new Green industrial Deal and new net-zero marine fuels.

T.6, (11:15-11:30). Thanapol Poojitganont, German Aerospace Center ..... Page 50.
Enabling Fuel Flexible Solid Oxide Fuel Cells (SOFCs) in Maritime Applications: 1D Simulation of SOFC System.

T.7, (11:30-11:45). Travis Teske, German Aerospace Center ..... Page 60.
Potential for Hybrid Propulsion Systems for a selected fleet in Germany.

T.8, (11:45-11:55). Alistair Aranha, German Aerospace Center ..... Page 6.
Quantification of resilience of concepts for short sea shipping of green hydrogen.

Session 10, (13:00-15:00) - Friday (M1)
Wind Assisted & Alternative Propulsion
Chair: Professor Emeritus Poul Andersen

T.1, (13:00-13:10). Jan-Erik Giering, German Aerospace Center ..... Page 25.
Model Predictive Control for Adaptive Power Management under Uncertainty in Hybrid Electric Ships.

T.2, (13:10-13:20). Aurore Wendling, (NTNU) ..... Page 68.
Early-stage PhD presentation: Wind as primary ship propulsion.

T.3, (13:30-13:45). Erik Blackert, KTH Royal Institute of Technology ..... Page 14.
Optimizing Sail and Vessel Design for Wind-Assisted Commercial Shipping: Trimming Strategies, Aerodynamics, and Sail Placement.

T.4, (13:45-14:00). Antonia Hillenbrand, KTH Royal Institute of Technology ..... Page 27.
Wingsail inflow at sea: Comparison of wind lidar measurements and CFD on an ocean-going vessel.

T.5, (14:00-14:15). Heng Zhu, Chalmers University of Technology ..... Page 77.
High-Fidelity Numerical Analysis of Hull-Wingsail Interaction.

T.6, (14:15-14:30). Chiara Wielgosz, RISE Research Institutes of Sweden ..... Page 69.
The Atmospheric Boundary Layer and its modelling in CFD for Ship Wind Propulsion Applications.

T.7, (14:30-14:45). Hauwa Yunusa, KTH Royal Institute of Technology ..... Page 73.
Future Ship Design For Decarbonization.

T.8, (14:45-15:00). Karolina Malmek, RISE Research Inst. of Sweden ..... Page 42.
Innovative Lifting-Line Methods for Wind Propulsion: Accounting for Sail-Hull Dynamics.

ON THE DESIGN OF BRIDGES AGAINST SHIP COLLISIONS

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A serious ship-bridge collision accident happens about once a year. These accidents cause fatalities and large economic losses due to loss of transportation service and replacement cost of the bridge structure. One of the most recent, widely published, ship-bridge collisions was the collision where the containership Dali in 2024 collided with the Baltimore Key Bridge in the US city of Baltimore. The resulting collapse of the bridge girder caused six fatalities as well as large financial losses. One effect of this event has been that engineers around the world now are being engaged in evaluation of the vulnerability of existing bridges and establishment of rational design criteria for new bridges.

The presentation will outline elements of a rational design procedure for bridge structures against ship collision impacts. A set of risk acceptance criteria will be proposed and a mathematically based procedure for calculation of the probability of ship collision accidents caused by human as well as technical errors will be presented. This first part of the presentation leads to identification of the largest striking ship, “design vessel”, a given bridge element must withstand without structural failure in order for the bridge connection to fulfil the risk acceptance criteria.

The final part of the presentation will be devoted to an analysis of the needed impact capacity for the bridge pylons and piers exposed to ship bow impact loads from design vessels. A procedure will be described for derivation of expressions for ship bow crushing forces, which can be used in design against ship collision impacts. The resulting collision force expressions are verified by comparison with large-scale laboratory experiments and an analysis of a full-scale shipping accident. Finally, the proposed impact force expressions will be compared with existing standards for modelling ship collisions against bridges as published by AASHTO, IABSE and Eurocode.

Marine Transport of CO_2 in Solid Form

Henrik O. Madsen[1]

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Abstract

Scenarios for reduction of greenhouse gas emissions towards 2050 to reach the Paris climate goals all include an element of Carbon Capture, Utilization and Storage (CCUS) of 6 billion tons per year or more. This means that minimum 6 billion tons of CO_2 must be captured, transported and utilized or stored every year. For comparison the world consumption of coal was 8.5 billion tons in 2023.

CO_2 can exist in four forms: gas form, supercritical fluid form, liquid form and solid form (dry ice). It can also be transported in each of these four forms. Transport in gas or supercritical fluid form takes places by pipelines, transport in fluid form in cryogenic, pressurized tanks, and dry ice is transported in boxes in pellet form although today only in small quantities.

The paper discusses bulk transport of CO_2 at sea in solid form as dry ice pellets. The pellets are assumed to be transported in insulated standard 20 ft containers, but bulk transport in cargo holds can also be envisaged.

Compared to marine transport in liquid form by liquefied CO_2 tankers, the transport by containers at ambient pressure has several advantages: it is almost intrinsically safe as it takes place at ambient pressure; it uses an existing transport chain both in terms of ships, terminals and containers, it is easy to scale and simple, and it is CAPEX light.

Safety issues around transport of dry ice in insulated containers will be particularly discussed.

Exploring Regenerative Business Models for Biodiversity Restoration in the Shipping Industry

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Abstract

The shipping industry has historically contributed to oceanic environmental degradation through practices that affect marine biodiversity. The impacts of shipping activities in aquatic and nearshore environments are varied and extensive, and can be divided into discharges to water, physical impacts, and air emissions. While traditional sustainability measures focus on reducing harm, they often fall short in addressing the urgent need for restoration and regeneration of marine ecosystems.

This paper investigates the feasibility of regenerative business models within the shipping sector, aiming not only to mitigate ecological impact but to actively contribute to biodiversity restoration in marine environments. By reviewing the literature at the intersection between biodiversity preservation/regeneration and shipping, the paper examines strategies, such as carbon capture technologies, oceanic habitat restoration partnerships, eco-friendly vessel design, ballast water management systems, antifouling practices, and biodiversity-supportive shipping routes.

I provide a mixed-methods analysis of industry practices and pilot programs to assess the economic and ecological viability of these approaches, exploring how shipping companies can transition from harm-reduction to positive ecosystem contributions.

The findings highlight the potential for the shipping industry to play a bigger role in reversing biodiversity loss, offering practical guidelines for integrating regenerative principles into maritime transport. The study identifies the operational shifts required for a regenerative approach and contributes to sustainable maritime operations by reframing the shipping industry as a potential agent of environmental renewal rather than merely a mitigator of damage.

Key words: Biodiversity; ocean shipping; regenerative business models.

Modeling autonomus shipping diffusion: an agent-based modelling

Dhaneswara Al Amien[1], Roberto Rivas Hermann[1] and Terje Andreas Mathisen[1]

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Abstract

Despite the emerging trend of modeling the diffusion of shipping innovations, current literature remains focused primarily on fuel-saving innovations. However, other emerging innovations in shipping, such as autonomous shipping, have gained some research and industry interest. Investing in autonomous shipping is a complex process involving both social influence and investment decision-making. This article aims to contribute to the literature on modeling the diffusion of autonomous shipping. Specifically, we examine how various factors, including policy, technology, and market dynamics, influence adoption rates. This paper uses agent-based modeling to simulate the diffusion of autonomous shipping among European dry bulk inland shipowners. While other research in shipping focuses on fuel price developments, this paper uniquely investigates crew wages, offering a novel perspective to the literature. The simulation results showcase various possible scenario outcomes and propose conditions for a broader adoption of autonomous shipping by inland shipowners.

Wind-Wave Coupling for Improved Modelling of Coastal Flooding

Gabor Albrect[1], Claes Eskilsson[1], Mads Røge Eldrup[1]
and Thomas Lykke Andersen[1]

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Abstract

Coastal flooding is expected to become more pronounced in Europe with the increased storm intensity caused by climate change. Today, wave induced coastal flooding modelling is typically carried out using shallow water models where the wind forcing is included through a surface drag term described by an empirical drag coefficient and the estimated wind velocity at 10 m height[1]. The drag coefficient is thus a critical factor and will heavily influence the results. The just started MSCA-PhD project aims to investigate the modelling fidelity of the atmospheric forcing and its influence on coastal flooding. This will be achieved using a coupled wind-wave model, thus completely circumventing the need of surface drag coefficient[2]. By coupling a coastal wave model to an incompressible Navier-Stokes model describing the atmospheric flow, the wind forcing can be directly given by the pressure working on the free surface.

In the presentation we will introduce the wind-wave coupling presently implemented in the spectral/*hp* element framework Nektar++[3]. To keep the numerical efficiency high, we work with static meshes and solve the fully nonlinear potential flow equations in sigma-transformed coordinates[4] as well as solving the incompressible Navier-Stokes in generalized coordinates[5]. The code coupling is achieved by MPI using the CWIPI framework[6].

[1] G. Wu, F. Shi, J.T. Kirby, B. Liang and J. Shi (2018): Modeling wave effects on storm surge and coastal inundation. *Coastal Engineering*, Vol. 140, pp. 371-382.
[2] G. Deskos, J.C. Lee, C. Draxl and M.A. Sprague (2021): Review of wind–wave coupling models for large-eddy simulation of the marine atmospheric boundary layer. *Journal of the Atmospheric Sciences*, Vol. 78(10), pp. 3025-3045.
[3] C.D. Cantwell, D. Moxey, A. Comerford, A. Bolis, G. Rocco, G. Mengaldo, D. De Grazia, S. Yakovlev, J.-E. Lombard, D. Ekelschot, B. Jordi, H. Xu, Y. Mohamied, C. Eskilsson, B. Nelson, P. Vos, C. Biotto, R.M. Kirby, and S.J. Sherwin (2015): Nektar++: An open-source spectral/hp element framework. *Computer Physics Communications*, Vol. 192, pp. 205–219.
[4] C. Eskilsson and A.P. Engsig-Karup (2024): Water wave simulations using fully nonlinear potential flow: Spectral/*hp* element models implemented in Nektar++. In: Ould el Moctar and Lantermann (eds), *Proceedings of the 26th Numerical Towing Tank Symposium*, 23rd – 25th October 2024 Mülheim/Ruhr, Germany.
[5] D. Serson, J.R. Meneghini and S.J. Sherwin (2016): Velocity-correction schemes for the incompressible Navier–Stokes equations in general coordinate systems. *Journal of Computational Physics*, Vol 316, pp. 243-254.
[6] ONERA (2024). https://github.com/onera/cwipi

Structural Integrity and Crashworthiness of Ships Powered by Small Modular Reactors

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Abstract

The integration of Small Modular Reactors (SMRs) as a power source for marine vessels offers significant advantages, including reduced environmental impact and extended operational range. However, this integration introduces new safety challenges, particularly regarding crashworthiness and structural integrity in the event of collisions or groundings.

This research investigates proposed structural enhancements aimed at improving the crashworthiness of SMR-powered ships. A double hull side structure from existing literature serves as a baseline to evaluate the effectiveness of the proposed solutions in strengthening the ship's structural integrity. The Accidental Limit State (ALS) is assessed on a real ship structure modified with these solutions to determine its response to collision and grounding events. Subsequently, the Ultimate Limit State (ULS) is analyzed to assess the solutions' impact on overall structural integrity. Long-term factors, such as fatigue and corrosion, are planned to be studied to evaluate the durability of these enhancements.

The long-term goal is to contribute to the development of regulatory frameworks for SMR-powered ship structures and assess the applicability of these solutions to existing vessels. By addressing both crashworthiness and long-term structural resilience, this research supports the safe and sustainable implementation of nuclear-powered marine technology.

Durability and Performance of Offshore Wind Turbines under Fatigue Conditions

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Abstract

The objective of designing offshore structures is to guarantee their safety and reliability from both environmental and economic perspectives. Representative data and models must be utilised to assess loads and responses accurately. A comprehensive understanding of meteorological and oceanographic conditions is vital for the fatigue design of offshore wind turbines.

This study investigates the fatigue reliability of critical structural components in offshore wind turbines. It defines the Fatigue Limit State failure event and identifies the different sources of uncertainty that may influence failures. The primary focus is on the critical fatigue design parameters that impact fatigue reliability and their potential implications for offshore wind turbines' safe design and operation.

Quantification of resilience of concepts for short sea shipping of green hydrogen.

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Abstract

In a move to decarbonise energy-intensive industries such as, heavy industry, shipping and aviation, green hydrogen is expected to play a key role in the transition to clean renewable energy in the European market. To achieve this transition, a great deal of research is required to ascertain the investment required to develop the infrastructure to produce, store and transport this green hydrogen. The development and implementation of this new infrastructure comes with a need to understand its vulnerabilities. The ability of a system to withstand, adapt to, and recover from disruptions or disturbances while maintaining its core functions, structures, and identity is called its resilience. It is important to understand the resilience of such critical infrastructure to identify its susceptibility to a variety of threats.

In a previous study by Bergström et al. (2024)[1] a Discrete Event Simulation (DES) based approach is employed to derive concepts for short sea shipping of green hydrogen produced at an offshore wind farm. A simulation-based case study was performed on green hydrogen produced at an offshore site in the North Sea, transported by pipeline to a port-based intermediate storage facility and then transported onwards by ship. The energy and resource efficiency of the transportation by ship, of 4 different forms of hydrogen carriers was assessed. Building on the aforementioned work, in this study a method using Bayesian networks (BNs) is proposed to quantify the resilience of the system under threats to its normal functioning. Bayesian Belief Networks (BBNs) provide a framework for the quantification of the performance of the system. BBNs can deal with uncertainties and partial data and information. They can also include both qualitative and quantitative measures of system performance. The performance metric is then used to generate a function for the resilience index of the system. The study addresses various forms of disruptions to the system ranging from port closure to collisions between ships and anthropological threats such as hijacking, to evaluate their effects on the performance of the system and thus quantify its resilience index.

The study aims to offer insights into the quantification of the resilience of such critical infrastructure so as to help identify vulnerabilities in the system, fortify the system against them and thus ensure minimal impact to the performance of the system in the event of disruptions or threats. This will provide key stakeholders with recommendations for building sustainable and resilient energy systems that take advantage of renewable electricity and green hydrogen to drive the decarbonisation goals of the EU.

[1] Bergström, M.; Braun, M.; Schmid, F.; Niemi, A.; Skobiej, B.; Dave, Y. et al. (2024): Simulation-based evaluation of concepts for short sea shipping of green hydrogen. In *IMDC*. DOI: 10.59490/imdc.2024.898.

Systematic Analysis of the Effects of Scale and Draft Variation on Form Factor Using a Combined EFD/CFD Method

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Abstract

In recent years, numerous studies have focused on predicting the form factor of a ship and the resulting scale-related impacts, as the reliability of this estimate has been questioned. This study focuses on investigating the difference between model and full-scale CFD-based form factor computations using a newly established method. The method is called *2-k* method, and it shows improvements in the accuracy of form factor estimation, providing better results by applying the full-scale process, especially in the case of stern flow separation phenomena. Three benchmark hulls have been tested in this study. The research uncovers a remarkable dependency of form factor on grid resolution through scale factor variations, while the results show consistency with the ones from the literature. Systematic variation of transom submergence is also executed which presents consistency of form factor results in both model and full scale. Implementing the systematic analysis in scale and draft variations, the results show that the *2-k* method is considered a holistic and trustworthy method, and it can provide reliable prediction of form factor, even in cases of presence of recirculating flow region behind a submerged transom.

Tracking Vessels Using Multiple Extended Object Tracking

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Abstract

Situational awareness is essential for any navigator, whether a human captain or an autonomous system, as effective operation requires understanding the surrounding environment, including the movement of other vessels. For autonomous vessels, this means that algorithms are required to detect and track other vessels. Traditional tracking methods represent targets as single points, an assumption valid at larger distances but less so in confined waterways where vessel size is crucial. Extended object tracking relaxes this assumption to allow a target to generate several measurements. The additional information allows us to estimate the size and shape of the vessel in addition to its position and velocity. Extended object tracking can also be combined with multi-object tracking methods to track an unknown number of targets. In this presentation, we will share recent research where we developed a multiple extended object tracker using the Gaussian process extended object tracking method due to its ability to model arbitrary shapes. We applied this method to track vessels, and we demonstrated the method on both simulated and real LiDAR data [1]. Furthermore, we will also share our research on how Automated Identification System (AIS) data can be incorporated into these methods by fusing the LiDAR data with the AIS data [2]. We also show results obtained from a real-world dataset from the Albert Canal in Belgium where we collected data from several barges to demonstrate that incorporating AIS data into the tracker results in improved tracking performance [3].

References:

[1] M. Baerveldt, M. E. Lopez, and E. F. Brekke, “Extended target PMBM tracker with a Gaussian Process target model on LiDAR data,” presented at the 2023 26th International Conference on Information Fusion (Fusion), Jun. 2023, p. 8.

[2] A. G. Hem, M. Baerveldt, and E. Brekke, “PMBM Filtering with Fusion of Target-Provided and Exteroceptive Measurements: Applications to Maritime Point and Extended Object Tracking,” *IEEE Access*, vol. 11/2023, 2023, doi: 0.1109/ACCESS.2023.0322000.

[3] M. Baerveldt, J. Shuai, and E. F. Brekke, “Improved Fusion of AIS Data for Multiple Extended Object Tracking,” in *2024 27th International Conference on Information Fusion (FUSION)*, Jul. 2024, pp. 1–8. doi: 10.23919/FUSION59988.2024.10706459.

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Goal/Risk-based Design of Methanol-powered Ships

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Abstract

Due to its potential for significant net emission reductions and compatibility with existing fuel infrastructure, methanol is expected to play a crucial role in decarbonizing the maritime sector. IMO's goal-based Interim Guidelines for the Safety of Ships using Methyl/Ethyl Alcohol as Fuel (IGM) provide a basis for regulatory approval of methanol-powered ships. The IGM outlines essential safety systems and requirements while providing possible solution approaches and allowing for the consideration of alternative designs. However, it was observed by the authors that shipyards and design offices encounter significant challenges in the practical application of these guidelines. In order to address these issues, an analysis was conducted to assess the applicability of the IGM. Furthermore a structured analysis process is proposed for the identification of the primary engineering tasks inherent in the design of methanol-powered ships without a reference vessel. Based on the results of the analysis a structured process for the design of a methanol energy system is proposed. This approach aims to simplify the application of the IGM. The proposed design process is validated with an illustrative case study demonstrating the desing of the tank ventilation outlet.

Keywords: Ship Design, Ship Safety, System Integration, Methanol Energy System, Goal Based Design

Target: Simplify the application of the interim guideline

Development of an inline inspection process for automated fatigue strength assessment of welded steel joints with data driven methods

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Abstract

Fatigue strength is the key design parameter for more reliable and lightweight steel structures in maritime applications and can bring different advantages for ships and offshore structures. For example, the energy consumption of maritime transportation can be reduced with an increased payload due to a more lightweight ship structure, the risk of structural failure can be reduced due to a higher reliability against fatigue failure and the general life cycle of ships and offshore structures can be extended. Especially, the welded joints are considered to reduce the fatigue strength of those structures. This is attributed to the notched weld surface geometry and other phenomena like residual stresses or increased hardness value which are related to the welding process. It is well known that the weld surface geometry varies along the weld seam and due to that also the stress concentration. Measuring techniques like 3D laser scanning offer the possibility to capture these precisely and holistically along the weld seam. However, most commonly the nominal stress approach is used for fatigue strength assessment, which only considers the weld seam surface geometry globally. High safety margins are therefore required to ensure reliability, which on the other hand means that the potential of lightweight construction cannot be fully exploited. The Finite Element Method (FEM) can be used for a much more accurate fatigue strength assessment, but are computationally time-consuming and therefore practically not applicable for a holistic assessment of the weld seam. One possibility to overcome the lack of complexity of the nominal stress approach and the high computing times of FEM, is by using data driven methods based on Machine Learning (ML) algorithms. However, the success of data driven methods depends on the quality and the representation of the data. Therefore, the research focuses on the utilization of existing data and aims to bridge the gap between modern measurement techniques and the state of the art of ML algorithms to provide the basis for an automated fatigue assessment tool for inline inspection processes.

Research gaps towards achieving good environmental status in our oceans with respect to underwater noise

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Abstract

The EU has since 2022 decided to include continuous underwater radiated noise (URN), such as from shipping, in the descriptors for GES, good environmental status. Each nation in EU thus must evaluate this descriptor and enforce targets and measures to mitigate harmful levels of URN. The procedure for assessment can loosely be described that based on AIS data and simplified models for the ship URN, a sound propagation model is used to compute the level of URN in the different areas of our oceans. These levels are then compared with the descriptor to evaluate GES in the area.

There are however, naturally, several simplifications, approximations, and uncertainties in each step of this procedure, including interpreting the descriptor itself. This presentation will outline some of the gaps the research community will need to try to close in the coming years in order to improve both the evaluation process as well as to mitigate URN and improve the health of the oceans. This will include topics such as characterization of the sources (the type of sound emitted by a vessel), coupling to propagation computations, environmental properties affecting propagation (e.g. speed of sound in the water, coast line and bathymetry, bottom sediment), measure of received noise, and impact on the marine habitat.

Weighted measurement of ship navigation competence during loss of ship control

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Abstract

Maritime safety and operational efficiency rely heavily on skilled navigation. To effectively evaluate the acquisition of these critical skills by nautical students and trainees, an objective and precise assessment system is crucial. This study introduces a novel approach to assessing maritime navigation skills during critical scenarios involving loss of ship control. The research addresses the limitations of traditional subjective evaluation methods by developing a weighted scoring system for ship navigation performance. This system incorporates a comprehensive range of competence dimensions, covering both routine operations and emergencies, aligned with the standard set by the STCW code.

The methodology employs the Analytical Hierarchy Process (AHP) to establish a weighted scoring mechanism, prioritizing specific competencies based on their relative importance as determined by experienced maritime professionals. This approach ensures that crucial factors carry greater weight in the overall assessment of navigational competence, particularly in scenarios involving loss of ship control.

This study examines the loss of control situation which has been a major contributor to accidents from 2018 to 2022. By establishing well-defined evaluation criteria and benchmarks, the new method enhances assessment consistency and effectiveness. It facilitates equitable comparisons among different navigators and across various similar evaluation scenarios, contributing to a more transparent and objective appraisal of navigation skills under challenging conditions.

This research not only provide a more robust framework for evaluating maritime navigation competence but also offers insights into critical skills needed during loss of ship control. The findings have significant implications for maritime training programs, potentially leading to improved preparedness and safety in navigating crises at sea.

A Second-Order WENO-based Finite Volume Method for Flooding Simulations using Shallow Water Equations with Enhanced Bathymetry Resolution

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Inland waterways provide an environmentally friendly and often more cost-efficient method of transporting goods across Europe. However, according to the UNECE, these waterways are under-exploited, and they plan on expanding the transportation network to increase their attractiveness. The reliability of waterways is key to promoting their use; therefore, flood control and accurate predictions are of utmost importance.

Subgrid methods for the shallow water equations (SWEs) are ideal for modeling inland flooding. These methods use two superimposed meshes: a coarse mesh, where the equations are solved, and an underlying subgrid mesh, where bathymetry data are stored to enhance water volume calculations, friction evaluation, and the accuracy of wet and dry fronts. The subgrid method was pioneered by Casulli [1]; however, the method presented here is explicit and shock-capturing, similar to the more recent approach by Sander et al. [2]. The method by Sanders relies on a cartesian grid, which limits geometric flexibility to handle accurately details of the terrain.

In this work, we present a new subgrid method for solving SWEs on an unstructured simplex mesh with an underlying unstructured subgrid mesh, offering increased flexibility. The model solves the SWEs using a Godunov-type method with an intercell approximate Riemann solver, making it shock-capturing. It employs a second-order spatial WENO-type finite volume method, based on slope-limiting techniques introduced by [3]. This flexibility and spatial accuracy make the method ideal for modeling floods near rivers and channels, where it effectively resolves high velocities. At the workshop, we will discuss the model's concepts and showcase several smaller proof-of-concept examples.

References:

[1] Casulli, V. (2009). A high-resolution wetting and drying algorithm for free-surface hydrodynamics. International Journal for Numerical Methods in Fluids 60 (4), 391-408.

[2] Sanders, B. F. and Schubert, J. E. (2019). PRIMo: Parallel raster inundation model. Advances in Water Resources 126, 79-95.

[3] Dumbser, M. and Käser, M. (2007). Arbitrary high order non-oscillatory finite volume schemes on unstructured meshes for linear hyperbolic systems. Journal of Computational Physics 221.2, 693-723.

Optimizing Sail and Vessel Design for Wind-Assisted Commercial Shipping: Trimming Strategies, Aerodynamics, and Sail Placement

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Abstract

My research focuses on advancing the design methodologies for the next generation of wind-assisted commercial vessels. The aim is to create knowledge that can optimize vessel performance through a combination of wind tunnel testing, CFD analyses, free sailing models, and full-scale experiments.

Key research areas include investigating trimming strategies for multiple sails under both static and dynamic wind conditions, with the goal of maximizing forward thrust and maintaining constant yaw moment. This involves wind tunnel testing and CFD analysis, as well as testing sail-course interactions using a 7-meter model in dynamic wind conditions.

Another area of interest is improving the aerodynamic performance of the vessel's hull and superstructure. By minimizing air resistance through optimized design, the research will explore different hull and superstructure shapes in wind tunnel tests, both with and without wind propulsion units.

Lastly, the optimal placement of multiple sails on the vessel will be studied, evaluating different sail positions and trimming for best forward thrust at various angles of attack. This will be assessed through wind tunnel tests and CFD simulations, providing insight into how sail arrangement impacts overall sailing performance.

Developing a Framework to Assess and Enhance Port Preparedness for Extreme Weather Events

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Abstract

Ports are vital nodes in the global supply chain, handling over 80% of the world's trade by volume. However, the increasing frequency and intensity of extreme weather events-ranging from flooding, hurricanes, and storm surges to heatwaves and sea-level rise-pose significant risks to the continuity of their operations. The disruptions caused by such events not only threaten port operations but also jeopardize global economic stability.

This research aims to develop a framework to assess and enhance the preparedness of ports in withstanding the impacts of extreme weather events. The study focuses on strengthening port infrastructure resilience, improving operational preparedness, and recommending necessary updates to policies that address climate risks in port environments. Key case studies of European ports are used to conduct a gap analysis of existing regulations to identify deficiencies in current preparedness measures.

The findings of this research will contribute to sustaining 80% of ports` operational capacity during extreme events and promote long-term economic stability by minimizing disruptions. The proposed framework and recommendations will not only enhance port safety but also contribute to broader climate adaptation strategies in coastal infrastructure.

This research addresses a critical need for ports to adapt to evolving climate threats. By developing a comprehensive framework, we ensure that ports are equipped with the capability to anticipate, prevent, respond to, recover from, and continuously adapt to the increasing frequency and severity of extreme weather events.

Keywords: Climate change, Extreme weather events, Ports, Mitigation, Adaptation, Preparedness.

Feedback on results vs feedback on behaviour. What works best for reducing consumption?

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Abstract

There is an increasing number of solutions within the “eco-feedback” genre, promising to reduce fuel consumption by offering the crew information that will hopefully change their behavior. Eco-feedback interfaces are not lacking in ambition, trying to incorporate learnings from behavioral science, persuasive technology and gamification.

Eco-feedback solutions often tap into the motivations and emotions of the user, as these are the main drivers for behavior. Dealing with humans and their emotions is not an easy task though, and the potential pitfalls are abound. Does the proud and autonomous captain really want someone to tell him how to control his vessel? Does the crew find it fair that their consumption is compared to other crews, that might sail under more favorable conditions?

The presentation will take a closer look at two distinct types of eco-feedback that are very common in the industry. *Feedback on results* is referring to the end result of the behavior (total consumption, consumption compared to other vessels etc), while *feedback on behavior* is telling how “eco-friendly” a specific action is (for instance if you accelerate too quickly).

Each of the two types of feedback comes with its own potential strengths and pitfalls, which designers, engineers and advisors working within the field should have knowledge of. Insights from sports psychology and pedagogy will be used to show the effect these different types of feedback have on learning and motivation, and what you should be aware of when designing and rolling out these solutions.

The presentation will also discuss the use of haptic levers as a mean for delivering direct feedback on behavior, and what kind of considerations should be made before implementation.

This presentation is the main topic of an ongoing PhD project in Ocean Industries Concept Lab.

The presenter holds a Major in Psychology from NTNU and is also an Industrial Designer from AHO. He has worked several years in the Maritime Industry as an interaction designer.

Cross-Disciplinary Insights for Maritime Safety: Applying Epidemiology and Biomechanics in Marine Craft Design and Operation

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Abstract

Marine craft are complex human-machine systems subjected to the stochastic nature of the sea. Occupants of these craft face challenges from the moving work environment and adverse conditions that reduce the overall performance of the human-machine system and jeopardize safety at sea. Sophisticated craft, tailored for demanding tasks, often fail to realize their potential as human operators grapple with various physical and cognitive impairments in dynamic marine environments. A balance between the craft and the human is required to optimize system performance. To achieve this, knowledge about human responses to working conditions aboard marine craft is essential yet lacking in the scientific community.

KTH Royal Institute of Technology, RISE Research Institutes of Sweden, and Dalarna University have partnered on a research project investigating how occupants aboard marine craft respond to their motions. The objective is to establish a scale linking the motion characteristics of marine craft to the risk of work-related psychophysical health issues and performance degradation among occupants. This scale can improve marine craft design and operations. The project intends to study spatial and temporal aspects in terms of whole-body kinematics, back and leg neuromuscular activity, and physical fatigue during exposure to authentic work environments at sea.

Epidemiology plays a pivotal role in identifying health risk patterns among various populations. Through epidemiological analysis, researchers can identify specific risk factors associated with prolonged exposure to vessel movements, including fatigue, musculoskeletal strain, and the cumulative effects of vibration on long-term health. These insights allow for targeted interventions, predictive safety measures, and strategies to reduce health risks and support the well-being of maritime personnel.

Complementing epidemiology, biomechanics provides a critical understanding of how the human body responds to shock, vibration, and motion at sea. By applying principles from mechanics, anatomy, and physiology, researchers analyze how tissues, joints, and muscles respond to external forces. Advanced modelling and experimental approaches help map shock wave transmission through body structures, identify vulnerable regions prone to injury, and develop protective solutions.

Together, epidemiology and biomechanics offer a comprehensive framework to enhance safety, human performance, and ergonomic standards in maritime operations. This interdisciplinary approach deepens the understanding of human responses to maritime conditions, promoting safer and more efficient craft design and operational protocols. This presentation will discuss how knowledge transfer from these disciplines can address specific challenges in the maritime domain, offering innovative solutions for improving the health and safety of marine craft occupants.

On the safety comparison of bunkering LH_2 and LNG

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Abstract

In order to reduce greenhouse gas emissions from shipping, which account for 3% of global emissions, new fuel must be introduced to the shipping industry. Hydrogen represents a potential fuel source for maritime transportation, in addition to methanol and ammonia. Liquified hydrogen (LH_2) offers superior storage densities, but also presents greater challenges due to its temperature of 20 K. Despite the fact that liquified natural gas (LNG), a cryogenic liquid, is already safely utilized in shipping, hydrogen is still approximately 90 K colder than LNG.

One particularly crucial phase is the bunkering, or the refueling of the ship. The cryogenic liquid must be transferred from one ship to another or from the shore to the ship in question via a flexible pipe or loading arm. It is common for concerns regarding the safety of bunkering cryogenic liquids to be raised. In previous work, the authors employed a quantitative risk assessment to compare the safety of bunkering liquid hydrogen to LNG, which is already safely used in shipping. The same system was assumed to be employed for both fuels. Due to the lower density of LH_2, one bunkering operation took five times longer than those of LNG, resulting in a fivefold increase in the frequency of leaks for LH_2. The higher leakage rate also resulted in significantly higher probabilities for hazardous events.

The consequences of the three hazardous events—explosion, flash fire, and pool fire—were also modeled, and safety distances for these events were identified. The results were more similar in this case. LNG required higher safety distances for a pool fire and an explosion, whereas LH_2 required higher safety distances for a flash fire.

As the bunker duration has been identified as the primary contributor to the discrepancy in the probabilities of the hazardous events, this work expands the system utilized for LH_2 from a 150 mm diameter to a 350 mm diameter, maintaining a comparable duration. The methodology employed in the previous work is then applied. The leak frequency of the hydrogen system is now significantly lower than that of the LNG system. The larger pipes and equipment are less prone to leakage, thus increasing the size decreases the frequency of leaks. The leak frequencies are used to calculate the probabilities of the three hazardous events—explosion, flash fire, and pool fire. Here, the lower leak frequencies of the LH_2 system are counteracted by the higher reactivity and thus the higher ignition probability of hydrogen. However, when comparing leak sizes of the same magnitude, the hazardous events have consistently higher probabilities for LNG than for LH_2.

If the leak size remains the same as in previous work, the safety distances remain the same as well. However, since the LH_2 system has been increased, a full rupture case is considered and requires higher safety distances than any other leak diameter. This can be easily explained by the fact that a larger amount of hydrogen is released.

In conclusion, it can be demonstrated that enlarging the LH_2 system reduces the probability of a hazardous event for hydrogen, while simultaneously increasing its safety distances due to the full rupture case.

Numerical Modelling of 3D gap resonances between side-by-side barges under combined wave-current excitation

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Abstract

Ship-to-ship (STS) transfer of clean fuels, e.g. LNG, methanol and ammonia, has been commonly used to ensure high bunkering volumes and allow for a stable supply of those fuels to market. Considering that bunkering operations usually take place in sea areas not far from the coast, side-by-side (SBS) configuration is a preferred solution as it does not require upgrades of carriers equipped with mid-ship transfer arrangements. During SBS operations, fluid resonances may occur when the primary wave frequency approaches one of the natural frequencies of the gap flow, which is a critical consideration in the design and operation of offshore structures. Besides, the existence of current modulates the wavelength to a large extent, especially at higher current speed, which has the potential to significantly influence gap resonances compared to wave-only conditions [1]. Additionally, the predominant focus regarding gap resonances has been on piston-mode responses [2-3], neglecting other responses observed in real SBS configurations of two ships. Therefore, a three-dimensional (3D) fully nonlinear numerical wave tank (NWT) is built up in the commercial CFD package STAR-CCM+ based on the Unsteady Reynolds-Averaged Navier-Stokes (URANS) equations, coupled with the Volume of Fluid (VOF) method. User-defined wave elevation and velocity, theoretically obtained from the stream function method, are implemented near the boundaries of the NWT to drive the flow field in the CFD domain through a one-way coupling process that modifies the fluid momentum in the forcing zones. Waves on uniform currents with a range of speeds are examined, considering both beam- and head-sea scenarios. The free-surface elevation and the spatial structure in the gap are obtained and compared with the wave-only cases. Based on the present study, the free-surface elevation is larger at a higher current speed under beam-sea conditions and the maximum response occurs at the middle of the gap, which should be treated seriously during the bunkering operation. On the contrary, the situation with waves and a small-speed current under head-sea conditions yields the largest gap response at a position near the bow within the gap of the side-by-side vessels. Therefore, looking at the gap response amid the gap is not conservative.

References

[1] **Ding Y.**, Walther J. H., Shao Y. (2024). Effect of wave-current interaction on gap resonance between side-by-side barges. *Applied Ocean Research*, 150: 104073.
[2] **Ding Y.**, Walther J. H., Shao Y. (2022). Higher-order gap resonance between two identical fixed barges: A study on the effect of water depth. *Physics of Fluids*, 34(5).
[3] **Ding Y.**, Walther J. H., Shao Y. (2022). Higher-order gap resonance and heave response of two side-by-side barges under Stokes and cnoidal waves. *Ocean Engineering*, 266: 112835.

Navigating uncertainty: The role of size and tie strength in shaping resilience for multinational corporations

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Abstract

Ongoing geopolitical distress challenges the operandi modi of multinational corporations (MNCs). To safeguard supply chains, MNCs implement resilience strategies to mitigate risks. While prior work in management science has primarily addressed standalone MNC resilience, little is known about how underlying market dynamics shape strategy implementation. Our study addresses this knowledge gap by integrating a meso-economic and organizational ecology perspective, proposing that supplier size and tie strength are critical in strengthening resilience. Capitalizing on a unique dataset that covers the entire U.S. import market and the World Uncertainty Index, our study examines global, subsidiary-to-subsidiary containerized trades by MNCs from 2021 to 2023. Supporting organizational theory, the results show that uncertainty negatively impacts MNC resilience, but sourcing capacity through larger and more strongly tied suppliers mitigates these adverse effects. By addressing MNC strategy implementation, the study broadens understanding of interdependencies within global supply chains that underpin resilience. From a managerial perspective, the study guides MNCs to strengthen ties with larger suppliers, which may inadvertently lead to higher supplier concentration in markets marked by uncertainty.

Data Utilization for In-situ Condition-Based Maintenance of Working Vessels in Offshore Wind

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Abstract

Nowadays, with high-powered computers becoming increasingly accessible and affordable, data analytics has become more feasible. This advancement has accelerated the digitalization of various industries, making it easier to manage and analyze large sensor-based databases. This has enabled multiple beneficial optimizations such as fuel consumption for a working vessel [1], time optimization of tugboat operations [2], and damage detection of ship hulls [3]. Despite these developments, it is widely recognized that several sectors within the maritime industry still lag behind other industries in adopting digitalization [4]. This lag is potentially attributed to a lack of standardization, which often results in large, unstructured databases with inconsistent quality and sources, making them difficult to access and utilize effectively, as recognized for fishing vessels [5].

While industries such as offshore wind have embraced digitalization for many years [6], the working vessels used for transporting personnel and equipment to these offshore wind farms have not yet embraced the benefits of digitalization. Here, a variety of data may be stored, yet further analysis is often lacking; in some cases, sensors have even been removed because the data was not used for any purpose. However, given that these vessels operate in demanding environments with a critical need for continuous operation, corrective maintenance is often prioritized, as preventive maintenance can be complicated by the necessity of uninterrupted operations. Thus, utilizing data for in-situ assessments of vessel conditions has the potential to enable an optimal maintenance strategy while ensuring continuous operation is maintained.

The objective of this presentation is to outline a structured approach for acquiring, processing, and utilizing data within a condition-based maintenance framework for working vessels in the offshore wind sector. This approach addresses the challenges of managing large, unstructured databases from multiple sources and effectively utilizing data for condition-based maintenance.

[1] J. C. Sørensen, M. Lützen, S. Eriksen, and J. B. Jensen, “A Modular Working Vessel Decision Support System for Fuel Consumption Reduction,” 2022, doi: 10.1142/S0219622022500109.
[2] S. Chen, F. Wang, X. Wei, Z. Tan, and H. Wang, “Analysis of Tugboat Activities using AIS Data for the Tianjin Port,” 2020, doi: 10.1177/0361198120916734.
[3] G. I. Aravanis, N. E. Silionis, and K. N. Anyfantis, “Damage detection in ship hull structures under operational variability through strain sensing,” 2023, doi: 10.1016/j.oceaneng.2023.115537.
[4] Z. Y. Tay, J. Hadi, F. Chow, D. J. Loh, and D. Konovessis, “Big Data Analytics and Machine Learning of Harbour Craft Vessels to Achieve Fuel Efficiency: A Review,” 2021, doi: 10.3390/jmse9121351.
[5] Z. Uriondo, J. A. Fernandes-Salvador, K.-J. Reite, I. Quincoces, and K. Pazouki, “Toward Digitalization of Fishing Vessels to Achieve Higher Environmental and Economic Sustainability,” 2024, doi: 10.1021/acsenvironau.3c00013.
[6] Z. Ren, A. S. Verma, Y. Li, J. J. E. Teuwen, and Z. Jiang, “Offshore wind turbine operations and maintenance: A state-of-the-art review,” 2021, doi: 10.1016/j.rser.2021.110886.

Use of Machine Learning for Estimation of Wave Added Resistance and its Application in Ship Performance Analysis

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Abstract

This study presents a deep neural network (DNN) model to estimate wave-added resistance using ships main particulars like length, beam, draft, block coefficient, and slenderness ratio. The training dataset is generated from strip-theory calculations across various hull geometries and operational conditions. The model provides fast and reliable wave resistance predictions, outperforming simplified empirical methods which often used in the industry for evaluation of ship performance. The DNN model delivers robust results when validated against direct numerical calculations from the DTU Solver. Additionally, we apply the model within a vessel performance framework to predict propulsion power and analyze biofouling for Ultrabulk ships.

Key factors like hyperparameter tuning and dataset selection greatly impact the performance of deep learning models. In this research, the numerical calculations, rather than experimental data, form the training dataset, allowing a more generalized and versatile approach. By optimizing learning rates and other parameters, the model overcomes issues like local minima. The addition of more hull data points and input features could further enhance the accuracy of the DNN model.

The model's integration into a ship performance framework enables operators to estimate delivered engine power and compare it with noon reports. This assists in evaluating hull and propeller efficiency, leading to more cost-effective voyages and energy-efficient ship operations.

The developed tool has potential for further refinement into a more accurate ship performance monitoring system. This system could improve ship performance through maintenance optimization and quantify the impact of energy-saving retrofits or advanced anti-fouling coatings. These efforts directly reduce greenhouse gas emissions and improve fleet environmental ratings, such as the Energy Efficiency Operational Indicator (EEOI) and Carbon Intensity Indicator (CII).

Hybrid WNPF-LSTM Modelling of Wave-Body Problems

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Abstract

Numerical models based on weakly nonlinear potential flow (WNPF) remain the tool-of-the-trade for wave-body interaction problems in Oil & Gas as well as for emerging technologies like floating offshore wind turbines (FOWT) and wave energy converters (WECs). Here a Morison-type drag is typically used to include nonlinear response into the models. By replacing the Morison-type nonlinear term with a recurrent neural network of long short-term memory (LSTM) type[1], thus introducing a hybrid WNPF-LSTM model[2], we can enhance the nonlinear performance of the models. This approach has been included in the WNPF model MoodyMarine[3], by coupling to the deep learning toolbox in Matlab[4]. We simply add a correction term δ to the Cummins equation. The correction term is made up by the LSTM prediction of the differences between the pressure and viscous forces obtained by a RANS simulation and the pressure force obtained by the WNPF having exactly the same motion trajectory, see Figure 1a. A key issue here is to minimize the amount of RANS training data which is time consuming to generate. Figure 1b shows the results of a heave decay test of sphere in which the LSTM model has been trained using up-chirps.

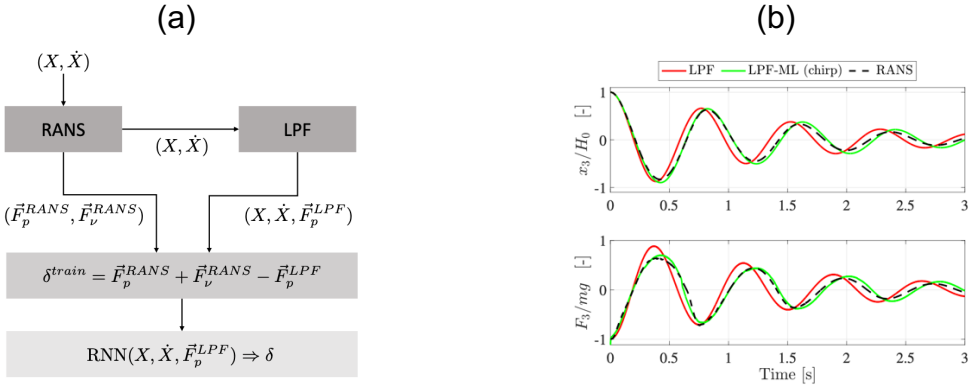


Figure 1: (a) training of correction term and (b) results for a heave-decay test.

[1] S. Hochreiter and J. Schmidhuber (1997): "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780.
[2] C. Eskilsson, S. Pashami, A. Holst and J. Palm (2023): A hybrid linear potential flow - machine learning model for enhanced prediction of WEC performance. In: *Proceedings of the 15th European Wave and Tidal Energy Conference*, Bilbao, Spain.
[3] J. Palm and C. Eskilsson (2023): Verification and validation of MoodyMarine: A free simulation tool for modelling moored MRE devices. In *Proceedings of the 15th European Wave and Tidal Energy Conference* European Wave and Tidal Energy Conference, Bilbao,Spain.
[4] Mathworks (2024): "https://se.mathworks.com/help/deeplearning/"

Experimental investigation of biomechanical response to shock and vibration exposure at high speed at sea

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Abstract

The work on board ships and boats is characterized by long shifts and physical vulnerability. The occupational health and safety legislation's action and limit values addressing vibrations and repeated shocks are regularly exceeded on board. What does that mean?

The presentation discusses an ongoing experimental campaign in the project *Working conditions and hull loads at high-speed operation*. The project aims to answer how the working environment at sea affects the risks to the crew, directly or as a result of months and years in service. The questions also concern how the working environment influences abilities such as concentration and decision-making.

The present fieldwork comprises, simultaneously measuring vessel motions and the crews' responses to these motions. The boat's hull and crews' seats are equipped with accelerometers and the crew members' motions, together with muscle activity in the neck, shoulders and lower back, are recorded. The goal is to quantify the relationship between the boat's acceleration and the body's response. The results are expected to link the motions of the ship to the risk that these will adversely affect the humans aboard.

Model Predictive Control for Adaptive Power Management under Uncertainty in Hybrid Electric Ships

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Abstract

The maritime transport sector is facing growing challenges as global pressure to reduce emissions and transition away from fossil fuels intensifies. This drive for defossilization introduces greater structural and technological complexities within modern energy systems, such as of Hybrid Electric Vessels (HES), where novel technologies like fuel cells and energy storage systems are becoming integral to their designs. At the same time, the growing unpredictability and intensity of weather conditions, driven by climate change, demand that vessels operate under increasingly uncertain environmental conditions.

These factors highlight the need for advanced Power Management Strategies (PMS) in HES that can efficiently and reliably maintain power balance during operation, while addressing both the uncertainties in environmental conditions and the decision-making process inherent in the strategy itself, as well as managing the structural and technological complexities of the energy system. Compared to more conventional approaches like rule-based PMS, real-time optimization-based strategies, such as those employing Model Predictive Control (MPC), offer greater potential to meet these requirements.

This “early-stage research” presentation outlines a PhD project aimed to investigate both the potential benefits and limitations of advanced PMS for HES based on MPC.

One main research aspect of the project focusses on the trade-off between decision-making effectiveness and required computational efficiency. This includes an assessment of specific functional aspects of different suitable MPC-based strategies, such as: (1) adaptability compared to non-adaptive strategies for addressing unexpected events and conditions (e.g., Adaptive MPC); (2) probabilistic versus deterministic modeling to manage uncertainties and errors in the decision-making process (e.g., Stochastic MPC); (3) nonlinear versus linearized modeling of complex dynamic system behavior (e.g., Nonlinear MPC), as well as (4) combinations of such approaches, potentially leading to PMS based on Adaptive Stochastic Nonlinear MPC.

The anticipated outcomes of this research will potentially include, among other aspects, a qualitative and quantitative characterization of selected PMS, along with a comparison of their performance under representative scenarios of distinct complexity to identify effects on their effectiveness and computational power demands.

Machine Learning for Ship Hull Design

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Abstract

The 2023 revision of the International Maritime Organization's (IMO) greenhouse gas (GHG) strategy has introduced stricter standards, underscoring the need to improve energy efficiency in the shipping sector. Key initiatives like the Existing Ship Energy Efficiency Index (EEXI) and Carbon Intensity Indicator (CII), now mandatory, require shipowners and operators to monitor and improve the environmental performance of their vessels. These regulations push industry toward solutions to reduce emissions and meet global sustainability goals.

One way to reduce ship resistance in the water and improve energy efficiency is through hull optimization. This, together with the increasing use of wind-assisted propulsion systems such as wing sails, offers a practical approach to meeting the IMO's ambitious goals. Emerging methods, particularly in the field of hull optimization, have increasingly turned to digital technologies such as Machine Learning (ML) and generative AI. These tools allow the exploration of a vast number of design variations and can significantly streamline the process of identifying optimal hull shapes, resulting in reduced drag and improved fuel efficiency.

According to the DNV Maritime Forecast 2050, the use of digitization, including AI-based optimization platforms, will be critical to meeting the evolving regulatory framework for maritime operations. ML, in particular, has shown great potential in accelerating ship design processes, reducing computational costs and exploring a wide range of configurations. This highlights the importance of developing ML and AI methods to anticipate regulatory changes and contribute to the decarbonization efforts of the shipbuilding industry.

The purpose of this presentation is to introduce my PhD thesis, which focuses on the application of ML- and AI-based techniques for hull design optimization. By integrating these methods into ship hull design, my research aims to support sustainable maritime practices and help the industry transition to a greener future.

Questions I want to address in the presentation:

- What should be the primary focus of the dataset needed for the training of the deep learning model?
- What strategies would most effectively promote widespread adoption of the AI ship design model within the maritime industry?

Wingsail inflow at sea: Comparison of wind lidar measurements and CFD on an ocean-going vessel

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Abstract

The need for decarbonization in the maritime industry leads to an exploration of wind as source of propulsion for ships. For harvesting the wind energy, different concepts of wind propulsion system are being developed. Common for all is the need for an automated operation. While the dimensioning at early design stages usually relies on wind statistics, an automation system needs to keep the wind propulsion system safe in whatever conditions the vessel is encountering. Recent standards for wind propulsion systems refer to onboard wind anemometer measurements for determination of the aerodynamic loads. However, the measurement of wind on a vessel is not trivial due to interaction effects with the hull and superstructure.
In this study, we compare onboard wind measurements with numerical simulations of the flow around a vessel in similar conditions. The wind measurement suite consisting of several anemometers at different locations and a wind lidar is installed on a Ro-Ro vessel which is proposed for retrofit of a wingsail. The wind lidar allows to assess the inflow conditions distributed over the span of a prospective wind propulsion device. In the complementary CFD simulations, the same vessel is investigated at different apparent wind angles. The inflow is modelled as superposition of an atmospheric boundary layer in the true wind (ITTC exponential profile with factor 1/9) and a reference vessel speed. This talk will explore the comparison of initial results in respect to their implications for the use in an automation system considering the weaknesses of the underlying methods.

Shipping fleet energy transition: Well to wake assessment model

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Abstract

Maritime shipping significantly contributes to global CO_2, SO_x, and NO_x emissions as a result of its dependence on fossil fuels. If the sector continues to rely on fossil fuels, emission is expected to increase with transportation demand and economic growth. Maritime transport at the EU level accounts for approximately 3 to 4% of the total CO_2 emissions, which amounts to over 124 million tonnes of CO_2 in 2021 [1]. Shipping is included in the EU's "Fit for 55" package, which aims to contribute to the EU's goal of carbon neutrality. One element in “Fit for 55” is the Fuel EU Maritime regulation which aims to decrease greenhouse gas emissions by reducing the yearly average GHG intensity of energy used on board a ship from a 2020 reference value of 91.16 gCO_2 equivalent per MJ, by 2% in 2025 and achieving the reduction target of 80% by 2050. Regarding energy transition, the shipping industry is commonly perceived as hard to abate due to numerous challenges, one of them being the low maturity and high cost of climate-neutral technologies.

Energy systems models are often used for a better understanding of long-term energy transition using a cost optimization mathematical model. Particularly to ships, energy systems models can also capture the political decision impacts, technological development, transport demand, and renewable energy availability to understand the shipping energy transition. Coupling life cycle assessment with an energy systems model can further enhance further understanding of the energy transition and its effect on the environment. However, coupling life cycle assessment with energy system modeling is challenging as both methods are data-intensive and complex. In this study through the integration of life cycle assessment into the global energy transition model, this study investigates well-to-wake greenhouse gas emissions and energy demand of low-carbon solutions of selected shipping fleets. The study shows more adoption of biofuels and e-fuels in shipping increases primary energy demand as the production of these fuels is energy intensive. Well-to-wake greenhouse gas emissions will continue to reduce with the assessed strategy, but more aggressive action will be needed to achieve a net-zero or 80% reduction target by 2050. However, findings indicate that the impact of fuel production pathways will become increasingly important for reducing carbon emissions in the shipping industry when adopting biofuel or e-fuels.

Liquid Hydrogen as a potential decarbonization solution for a large cruise ship

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Abstract

The global maritime industry was in 2022 responsible for over 2 % of CO_2 emissions and other air pollutants [1]. As a result of the growing environmental concerns, the need to decarbonize arose. The International Maritime Organization (IMO), acting as the international regulatory body, set goals in 2018 to achieve 50 % reduction in carbon emissions. As of July 2023, the target has been updated to reach net-zero greenhouse gas (GHG) emissions by 2050 [2]. To meet the decarbonization targets, it becomes necessary to use new energy sources enabling low or zero carbon emissions to substitute the currently used fossil fuels. The alternative fuel options range from low-carbon fossil fuels like liquefied natural gas (LNG) and methanol to carbon free fuels like hydrogen and ammonia. Moreover, the type of energy converters and systems onboard that can transform those alternative fuels into power has to be considered as they contribute to ship overall GHG emissions. Among different existing energy converter, fuel cells are efficient and clean options compared to combustion-based systems [3].

This study examined the energy transition possibilities of a sample vessel, which is a large cruise ship. The proposed decarbonization solutions are liquid hydrogen, coupled with proton exchange membrane fuel cell (PEMFC) and solid oxide fuel cell (SOFC) systems for onboard energy conversion. The focus is on these fuel cell types because of the availability of information supplied by technology providers. In the case of this cruise ship, a technical assessment has been made to compare the different scenarios based on criteria including fuel volume and weight requirement and availability of space on board; the size of the energy converter and annexed systems like batteries; the carbon emissions from the system in comparison to the use of fossil fuels. Such assessment will provide a better overview of liquid hydrogen as fuel onboard large cruise vessels and the findings of this work will serve as a base for decision-making on other cruise ships both retrofits and newbuilds. It is noteworthy that in addition to the technical assessment results, the decision-making is subject to a financial assessment to include the costs related to such energy transition.

[1] „International Shipping,“ 11 7 2023. [Online]. Available: https://www.iea.org/energy-system/transport/international-shipping.

[2] N. K. Kyokai, „Pathway to Zero-Emission in International Shipping,“ ClassNK, Tokyo, 2023.

[3] G. Guadagna, P. L. Christensen und K. Matsushita, „Fuel Cell Technologies and Applications for Deep-Sea Shipping,“ Maersk Mc-Kinney Moller Center for Zero Carbon Emissions, Copenhagen, 2024.

A novel method for maritime risk assessment based on first principles

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Abstract

To ensure the safe navigation of ships it is essential for the maritime authorities and relevant stakeholders to have an effective and reliable tool for periodic assessment of maritime traffic risk, particularly in areas where waterways cross or merge. In such areas with high traffic density there is a risk of potentially hazardous and undesirable events. One of them is ship collision. The consequence of such an event is not only financial losses, but sometimes also environmental pollution.
The framework of maritime risk assessment has been well developed, but most of the methods for quantitative risk assessment consider maritime traffic in a static or quasi-static manner. These methods focus on the association of accident risk with ships characteristics and traffic density. There is lack consideration of influence of traffic complexity, waterway complexity, and environmental and human factors on collision probability. To fill this gap the proposed novel method presents an approach for maritime water risk assessment based on human reliability, including elements which are the main drivers for human performance: traffic, waterway, and environmental complexity. For the purpose of defining factors human reliability modelling, big data analysis, and Bayesian Networks was employed. Also, parameters of the developed model are based on survey in maritime domain and knowledge elicitation from maritime experts. Model is applied to selected case studies to present first results of the novel method used.

Assessment of Blockage Effect on Cavitation Dynamics and Underwater Radiated Noise: A Numerical Study

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Abstract

Underwater radiated noise from ships has gained attention due to environmental concerns for the wellbeing of marine animals. Sea trials for underwater noise assessment still present significant challenges as they are conducted in uncontrolled environments. As a result, model-scale tests in cavitation tunnels are more commonly used for these measurements. However, the confined space of a typical cavitation tunnel introduces its own challenges for measuring underwater noise. One of those challenges is the blockage effect, caused by the small ratio of cross-sectional area between the tunnel and the ship model.

This work presents a numerical study on blockage effect. The test case involves a propeller operating behind a chemical tanker hull under cavitating flow conditions. Simulations are performed using a computational domain sized to match the RISE cavitation tunnel, while an additional simulation with a larger domain represents free-field conditions. The analysis focuses on the wake field, cavitation dynamics, pressure pulse levels, and noise emissions. All simulations are performed using the commercial software package Simcenter STAR-CCM+. The Reynolds-Averaged Navier-Stokes (RANS) approach is applied, with the Schnerr-Sauer mass transfer model for cavitation. The Ffowcs Williams-Hawkings acoustic analogy is used for noise levels assessment.

Can self-driving car rules streamline autonomous ship regulations?

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Abstract

The rapid advancement of autonomous technology is driving transformative change in various industries, with transportation being one of the primary sectors experiencing greater flexibility and independence as it transitions away from human-operated mobility systems. While it's clear that road and maritime domains are different, growing curiosity surrounds whether the regulatory frameworks developed for self-driving cars on land could pave the ways for their maritime counterparts.

This presentation and subsequent paper submission aim to discuss the parallels and divergences in technological complexity, operational environments and stakeholder ecosystems, and it analyzes the extent to which regulatory insights from autonomous vehicles could be applicable to the maritime domain, particularly in 6 areas: safety and liability, certification, cybersecurity, data privacy, ethical decision-making, traffic law and infrastructure adaptation.

Despite the perception requirements for ship navigation and maritime traffic management systems exhibit differences from those of autonomous land vehicles, the advanced control systems and automation technologies developed for conditionally and highly automated vehicles are anticipated to give a considerable positive influence on the development of autonomous shipping systems. The challenges and solutions that arise in road traffic AI regulations, and the cross-industry insights could accelerate the development of streamlined regulatory framework for enhanced multimodal transports and more scalable regulations for autonomous ships.

2nd-year results from the i-MASTER Horizon Europe Project

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Abstract

This presentation includes the research outcomes achieved during the second year of the i-MASTER Horizon Europe Project, which is a research and innovation initiative funded by the European Commission. The project focuses on integrating emerging technologies into maritime education and training. In the second year, research advancements have been made in maritime learning analytics and learning library, further solidifying its role within the project.

This presentation will highlight these key research progress and illustrate how cutting-edge technologies are transforming maritime education and presenting the potential benefits for the broader maritime education and training community. This overview will cover the main deliverables, the research challenges encountered, and outline current activities as well as future research directions. The presentation aims to foster public engagement, encourage research collaborations and also gather feedback to further refine and enhance our research approach.

By advancing more digitalized and efficient training methods, we aim to improve the overall effectiveness of maritime education, enhancing both learning outcomes, safety and operational efficiency in the industry.

The new Green industrial Deal and new net-zero marine fuels

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Abstract

Green fuel for shipping: Over recent years, there has been a considerable focus on the development of green fuels for shipping (PtX: methanol, ammonia). However, several Danish construction projects have encountered so many difficulties and delays that the mentioned initiatives have either been put on hold or await further development. Which initiatives, including governmental ones, can help start/restart the development?

Industrial policy: The framework conditions are clearly not sufficient and the green technologies not sufficiently competitive. There are expectations - also among Danish shipping companies - that the EU's industrial policy plan *»A Green Deal Industrial Plan for the Net-Zero Age«* can contribute to new solutions. The plan shall contribute to the green transition, secure EU employment, contribute to the EU's security of supply with a focus on regulatory simplification and investments in green technologies. The Green Deal Industrial Plan must be developed further, and there are hopes for a Danish fingerprint in the new Danish commissioner for the area. Based on this background, we will particularly focus on the following three topics:

1) The current development: The development of Danish marine fuel projects. Which projects have been formulated, which have been implemented, which projects await further clarification, divided into the various technology platforms. The project progress is assessed according to regulatory requirements, financing, fuel prices, local circumstances and other factors which have had an impact on the postponement or cancellation of the various construction projects.

2) Framework conditions: The current Danish and EU framework conditions and their possible improvements. What is needed and what is missing. The new EU Commission has announced that a new "Green Industrial Deal" must be formulated, which, among other things, must ensure access to cheap, sustainable and safe energy. The Shipping Association has recommended that the shipping industry should be put in the driver's seat for the development of the new "Clean industrial Deal". What elements should or can be added to the framework conditions that can create a Danish and European large-scale development of green marine fuels?

3) Cooperation and development: European production, employment and security of supply are at stake. Can an extended *Nordic cooperation* be established and create the necessary conditions (technology and framework) for the development of green fuel for shipping. Can experiences and perspectives from other countries be included in this collaboration, e.g. Maersk's Chinese agreements (the agreement with Goldwind for 0.5 million tons annually from 2026)?

What risks are covered with current safety practises in the Maritime business?

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Abstract

This PhD research aims to enhance risk management and safety practices within the maritime industry, focusing on traditional segments like tankers, bulk carriers, and passenger vessels. The study will assess current safety regimes, identifying challenges in their effectiveness in mitigating operational risks. By introducing and testing a real-time risk prevention and mitigation methodology, this research seeks to provide a more adaptive and human-centered approach to maritime safety. Leveraging interdisciplinary insights from social sciences, psychology, and risk science, the project will propose innovative frameworks that account for the complexities and uncertainties inherent in maritime operations. The ultimate goal is to develop a practical, adaptable tool that will foster more effective risk management and contribute to a safer maritime environment.

Heavy tails and high-order probability density functions for the surface elevation in irregular seas

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Abstract

As shown in the classic work of Longuet-Higgins (1963), the probability density function (PDF) for the free surface elevation in irregular seas has an integral formulation when based on the cumulant generating function. To leading order, the result is Gaussian, whereas nonlinear extensions have long been limited to Gram–Charlier series approximations. As shown recently by Fuhrman et al. (2023), however, the second-order integral can be represented exactly in closed form. The present work (Klahn et al. 2024) extends this further, enabling determination of this PDF to even higher orders. Towards this end, a new ordinary differential equation (ODE) governing the PDF is first derived. Asymptotic solutions in the limit of large surface elevation are then found. These provide new analytical forms for the positive tail of the PDF beyond second order, and also clarify how high-order cumulants (involving e.g. the kurtosis) govern the tail, which gets heavier at each successive order. The asymptotic solutions are finally utilized to generate boundary conditions, such that the governing ODE may be solved numerically, enabling novel determination of the PDF at third and higher order, without resorting to narrow-band or uni-directional assumptions. Successful comparisons with challenging data sets based on irregular wave theory (Madsen & Fuhrman 2012), numerical simulations (Klahn et al. 2021) and experiments (Trulsen et al. 2020) confirm accuracy. The methodology thus enables the PDF of the surface elevation to be determined numerically, and the asymptotic tail analytically, to any desired nonlinear order. Results are worked out explicitly up to fifth order. The theoretical probability of extreme surface elevations (typical of rogue waves) may thus be assessed quantitatively for highly nonlinear irregular seas, requiring only relevant statistical quantities as input. (This research has been supported by Independent Research Fund Denmark, grant ID 10.46540/2035-00064B.)

References

Fuhrman, D.R., Klahn, M. & Zhai, Y. (2023) A new probability density function for the surface elevation in irregular seas. *J. Fluid Mech.* **970**, A38.

Klahn, M., Madsen, P.A. & Fuhrman, D.R. (2021) Simulation of three-dimensional nonlinear water waves using a pseudospectral volumetric method with an artificial boundary condition. *Intl J. Numer. Meth. Fluids* **93**, 1843–1870.

Klahn, M., Zhai, Y. & Fuhrman, D.R. (2024) Heavy tails and probability density functions to any nonlinear order for the surface elevation in irregular seas. *J. Fluid Mech.* **985**, A35.

Longuet-Higgins, M.S. (1963) The effect of non-linearities on statistical distributions in the theory of sea waves. *J. Fluid Mech.* **17**, 459–480.

Madsen, P.A. & Fuhrman, D.R. (2012) Third-order theory for multi-directional irregular waves. *J. Fluid Mech.* **698**, 304–334.

Trulsen, K., Raustol, A., Jorde, S. & Rye, L.B. (2020) Extreme wave statistics of long-crested irregular waves over a shoal. *J. Fluid Mech.* **882**, R2.

CFD and experimental study of a model-scale two-body hinged raft wave energy converter

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Abstract

The two-body hinged raft wave energy converters (WECs) have a significant potential to harvest ocean wave energy for electricity generation. Optimization of these devices is crucial for efficient energy conversion, and computational fluid dynamics (CFD) simulation can serve as a pivotal tool to achieve this goal. CFD is capable of unveiling the underlying physics since it can simulate nonlinear waves and nonlinear wave-structure interaction. A validation of a CFD model for a 1:20 model-scale WEC from Crestwing will be shown in case of moderately nonlinear and nonlinear waves. Furthermore, a sensitivity study will be presented with variation of mass, position of the center of mass and the geometry of the device. This is the first step to explore the potential for optimizing the wave energy converter towards higher capture width ratio (CWR) and lower levelized cost of energy (LCOE).

Coastal green energy supply stations

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Abstract

This research project focuses on analyzing the sustainability of alternative maritime fuels and developing a supply network for bunker stations and green corridors for coastal shipping in Norway.

At the time being bunker stations and the supply of alternative maritime fuels are very limited in Norway's costal area. A reliable alternative maritime fuel infrastructure and supply chain is necessary to run the costal fleet in future on low emission to ensure to reach Norway's emission reduction goals for 2030 and 2050. Ship owners need predictable planning to convert their ships from fossil fuel to alternative fuels. This project aims to support this sustainable transition in costal shipping by providing necessary information for a green infrastructure.

The work is part of a PhD project. It is conducted as part of the REFUEL research project which is funded by the Norwegian research council and nine industrial partners. There will be an exchange of knowledge between the partners and the research project. To identify alternative maritime fuels an LCA will be conducted. A library of data on sailing routes and energy supply/demand will be collected to develop a network model.

The goal of the PhD project is to develop an energy supply network model for alternative maritime fuels based on investigations on future demand and supply structures. Furthermore, this will provide a basis for implementation of a reliable bunker supply chain network for alternative fuels, including guidance to ship owners on decisions on technical solutions and conversions of their ships. For potential bunker stations a design of land-based and floating arrangements will be provided, allowing a flexible up- and down-scaling depending on demand.

Finally, the aim of the project is to give recommendations to decision makers and government agencies to support policy decisions, and to support society and businesses to enable a sustainable maritime transition. The outcome of this project will be scalable and applicable to other regions and therefore provide a multiple added value and contribute to a zero-emission shipping environment.

Research at NTNU towards net zero shipping

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Abstract

This presentation illustrates present and future research initiatives towards net zero shipping emissions by 2050 at the Department of Marine Technology, NTNU Trondheim. In general, such initiatives and corresponding emission reduction measures may be broadly grouped into

- Logistics: Improve the utilization of ships
- Energy efficiency: Reduce energy consumption of all onboard systems
- Alternative fuels: Reduce the emission intensity of any additional energy (bunkered fuels or charged batteries)

As of today, no single silver bullet has been identified and thus the Department of Marine Technology conducts research on various emission reduction measures and technologies within each of the three groups. We outline the most recent results, and what we see as the most important topics for the future.

Physics-guided metamodel for vertical bending-induced fatigue damage monitoring in container vessels

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Abstract

A novel physics-guided metamodel will be presented to predict vertical bending-induced fatigue damage in a 2800TEU container vessel navigating the North Atlantic, based on data from the vessel's hull monitoring system. The metamodel combines two XGBoost-based base learners: a black-box model utilizing ship heave and pitch motion measurements, and a gray-box model using spectral moments from numerical analysis. Predictions from both models are refined through a meta learner Gaussian process regression to enhance accuracy. The metamodel was evaluated against black-box and gray-box models across various training data volumes. The proposed model adapts to varying data volumes, from months to over 2 years, effectively integrating the strengths of both base learners to provide reliable predictions in both seen and unseen scenarios. The model consistently demonstrated superior performance, enhancing fatigue damage accumulation accuracy by up to 35% over traditional machine learning methods. This advancement can aid the maritime industry in effectively monitoring ship fatigue and implementing predictive maintenance strategies, marking a significant step forward in applying data-driven techniques in shipping.

Influence of turbulence model selection on model-scale propeller and rudder performance

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Abstract

The propeller and the rudder are two key components of ship design for the ship's propulsion and maneuverability. The propeller operates in the wake of the hull and strongly influences the flow around the rudder. Similarly, the flow around the propeller is also affected by the presence of the rudder. However, ship performance, and consequently propeller performance, is typically evaluated at model-scale, whether relying on numerical approaches or on experimental data. In both cases, the propeller is commonly considered on its own, thus lacking the interaction effect with the rudder. The flow is often considered as turbulent, either by employing tripping in experiments, or carrying out the simulations as fully turbulent.

In this work, numerical simulations are carried out using STAR-CCM+ for a four-bladed propeller and rudder configuration at model-scale. The angle of attack of the rudder is varied while keeping the advance coefficient of the propeller constant. Different turbulence models such as the realizable $k-\epsilon$ and the $k-\omega$ SST models are employed in order to study the influence of the turbulence model on rudder stall and the performance of the rudder and propeller.

Innovative Lifting-Line Methods for Wind Propulsion: Accounting for Sail-Hull Dynamics

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Abstract

The (re)introduction of wind propulsion for ships calls for new efficient aerodynamic performance prediction tools within the field of naval architecture. In early design, there is a need for tools with low computational costs and easy adaptation to new sail configurations, yet with the ability to capture major flow effects with acceptable accuracy. Two important flow effects are the sail—sail flow interaction and the sail—hull/superstructure interaction. These interaction effects has a large impact on the sails' load distribution and stalling point, which directly impact the driving force generated by the system.

In recent years, several numerical versions of lifting-line methods, many based on Phillips and Snyder's Modern Adaptation of Prandtl's Classic Lifting-Line Theory, have been suggested for this purpose (Phillips, 2000; Malmek, 2023). In these methods, the three-dimensional flow effects are modeled using in-viscid vortex models while the two-dimensional profile lift and drag are obtained from viscous CFD simulations or wind tunnel data.

Different variations of lifting-line codes have been successfully applied to interacting wing sails, airplane wings, wind turbine blades, and hydrofoils. However, there are few examples where the effect of sail—hull/superstructure interaction has been accounted for. This paper presents a method for accounting for sail—hull/superstructure interaction in lifting-line codes. Two different hull and wingsail geometries are used for validation; a crescent-moon shape (Zhu, 2023) and a two-elements wing sail and hull of the RISE-SSPA Workshop On Performance Prediction for Wind-Powered Ships (SWOPP). The validation data is generated using computational fluid mechanics.

Phillips WF, Snyder DO. (2000). *Modern adaptation of Prandtl's classic lifting-line theory*. Journal of Aircraft 37(4):662–70. https://doi.org/10.2514/2.2649

Malmek, K., Larsson, L., Werner, S., Ringsberg, J. W., Bensow, R., & Finnsgård, C. (2024). *Rapid aerodynamic method for predicting the performance of interacting wing sails.* Ocean Engineering, 293, 116596. https://doi.org/https://doi.org/10.1016/j.oceaneng.2023.116596

Heng Zhu, Hua-Dong Yao, Fabian Thies, Jonas W. Ringsberg, Bengt Ramne. (2023). *Propulsive performance of a rigid wingsail with crescent-shaped profile.* Ocean Engineering Volume 285. https://doi.org/10.1016/j.oceaneng.2023.115349

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Evaluating Iconography Design Guidelines for Navigation Control Function Icons: Systemic Design Approach to Enhancing Usability in Integrated Navigation Systems

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Abstract

Icons within Graphical User Interfaces (GUIs) are objects used for visually identifying or emphasizing information that is either available textually and/or ecologically present in the human environment. These icons serve dual roles as they can invoke digital functions and open related objects when engaged with a cursor within an interactive system. Present on various digital and physical devices, such as a mobile phone and computer operating systems, icons are also fundamental components in interfaces of Integrated Navigation Systems (INS).

Currently, iconography design in INS interfaces is guided by the IEC 62288 guidelines, which serves as standards widely adopted by industry actors in INS development. However, past research indicates, that these icons have frequently failed to meet essential usability principles, particularly in terms of comprehensibility and distinctiveness. Additionally, the International Maritime Organization's (IMO) MSC.1/Circ.1609 - Guidelines for the Standardization of User Interface Design for Navigation Equipment - "S-mode guidelines" - aimed at promoting standardization, has yet to be effectively implemented. This raises questions about the real-world implications impacting the efficacy of these guidelines.

In absence of either the ability to meet the usability aspects or effectively adopt standardized guidelines that enhance human-centricity in interfaces, persistent concerns regarding human safety remains. This presentation provides an in-depth design analysis of existing IEC 62288 iconography used in current INS systems and offers a critical evaluation of these icons through a design perspective, including user and expert interviews, survey and reflections. The findings highlight design-related inadequacies identified in the IEC 62288 iconography guidelines and propose potential solutions for improvements. This study emphasizes the need for revisiting crucial guidelines that focus on human-centricity, open innovation-led standardization initiatives which may support bridging the gap between industry sustainability and user safety needs in maritime navigation, responding to the growing demands for usable e-navigation systems.

Sea State Estimation from Wave-Induced Ship Responses using Machine Learning

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Abstract

This study investigates the application of advanced data processing and machine learning techniques for determining the sea state encountered by a ship in a seaway. Measurements of the wave-induced responses on a large ship during transoceanic voyages are processed and utilized to derive estimates of the wave parameters and spectra in the experienced seaway.

The proposed methodology leverages various machine learning techniques, such as LightGBM and Artificial Neural Networks (ANN). The use of reanalysis data from freely-accessible public repositories – like ERA5 from the European Centre for Medium-Range Weather Forecasts (ECMWF) – is evaluated for its adequacy in model training and testing purposes. The preliminary results demonstrate the efficacy of machine learning techniques in enhancing the accuracy and reliability of sea state estimates.

Providing reliable, real-time information about the local wave conditions is crucial for ship operators and onboard decision-support systems, facilitating efficient decision-making and enhancing safety and fuel efficiency in maritime operations.

Non-intrusive Reduced Order Modelling for Free Surface Flows

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Abstract

This project aims to make a non-intrusive reduced order model (ROM) to solve nonlinear partial differential equations focusing on free surface flows. The motivation behind this is to obtain a numerical speedup compared to conventional numerical solvers without knowing the system's governing equations. This is useful, e.g., design optimization of parameterized problems. To obtain these properties, we propose a dimensionality reduction method and a time series regression method for the nonlinear latent dynamics of the shallow water equations.

A method to reduce the dimensionality of the system is the proper orthogonal decomposition (POD) [1]. The POD is a method that learns a small set of basis elements that describes most of the underlying dynamics of a data set via a singular value decomposition. The data set is a snapshot matrix of the solution created by either a numerical scheme or observed data. Because of the singular value decomposition properties, the generated basis transformation also has an inverse, which is why the POD can be used as an autoencoder.
Once a latent space is obtained, a suitable time regression method can be chosen. For non-intrusive purposes, this means selecting a data-driven method to learn the latent time series data, i.e., the latent transformation of the snapshot matrix. Considering the nonlinearity of free surface flows such as the shallow water equations, we propose using a Neural ODE [2], which is known to work well for learning nonlinear dynamics.

In this work, we use a POD to create an autoencoder and a Neural ODE to learn the nonlinear latent dynamics of the shallow water equations. The data is obtained using a spectral element-based solver, and the non-intrusive method will be compared to a spectral element model to analyze its numerical performance. Specifically, we will focus on the tradeoff between efficiency and accuracy.

References:

[1] S. L. Brunton and J. N. Kutz. *Data Driven Science and Engineering 2. edition*, Cambridge, 2022.
[2] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. *Neural Ordinary Differential Equations,* arXiv, 2019.

Steering new habits: How feedback can increase awareness of our energy usage. An integrated Systematic Review of Eco-feedback.

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Abstract

Feedback has been used in different disciplines, including interaction design, Human-computer interaction, and cybernetics, as well as in contexts such as maritime, aviation, and social science, to help people understand their behavior based on their experiences. This systematic review aims to understand the current landscape of feedback on energy consumption, specifically focusing on a strategy known as Eco-Feedback. Eco-feedback is designed to provide insights into individual or group behaviors, aiming to reduce environmental impact and encourage pro-environmental behavior. This strategy has been adopted in various industries, such as aviation, automotive, and construction, to decrease energy consumption. Research has shown that adopting an Eco-feedback approach can lead to significant energy savings by 5–20%.

In this paper, I reviewed 40 relevant studies published between 1980 and 2023 using the PRISMA protocol for systematic reviews. This paper represents the first comprehensive framework that encompasses various eco-feedback strategies or techniques that can assist practitioners, such as designers and researchers when designing or evaluating user interfaces for energy management. Following this review, I applied the proposed framework to analyze vehicle user interfaces. And offered recommendations for future work. Through this analysis, I developed a comprehensive design report explaining how various car manufacturers implement and display different eco-feedback strategies on car dashboards.

Keywords: Eco-feedback, user interface, behavior change, sustainability, interaction design, user experience, human-computer interaction.

Improving Energy Savings Onboard Passenger Ship

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Abstract

Improving energy efficiency onboard passenger ships is crucial for meeting the shipping industry's evolving sustainability goals and minimizing overall fuel consumption. Passenger ships are equipped with various onboard hotel systems, such as HVAC (Heating, Ventilation, and Air Conditioning), galleys, lighting and other amenities designed to enhance passenger comfort. These hotel operations significantly contribute to fuel consumption due to the dynamic power demand of the systems and their electrical power consumption is influenced by several factors, including operational and environmental conditions.

This study aims to investigate the influence of external factors, including passenger count, environmental conditions, and operational parameters, on the ship's largest electricity consumers within the hotel operations. Employing Random Forest regression and other statistical techniques, we analyze operational data from a Roll-on/Roll-off passenger (RoPax) ferry to identify the main drivers of energy consumption and develop targeted strategies for reducing energy use.

In addressing the research objectives, this study provides practical insights into optimizing energy-saving strategies onboard, contributing to the broader goal of improving energy efficiency in the maritime sector. The findings have implications for operational cost savings and compliance with evolving regulatory standards.

Keywords: Ship hotel systems, Electrical power consumption, Random Forest regression, Passenger ship

On GPU-acceleration of non-linear ship motion simulations in time-domain

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The IMO's Second Generation Intact Stability Criteria (SGISC) highlights the importance of addressing highly nonlinear phenomena, such as parametric roll and pure loss of stability in waves, through time-domain simulations (IMO, 2020). These simulations are crucial not only for Direct Stability Assessment (DSA) but also for Operational Guidance (OG). However, a significant challenge remains: the high computational demands, particularly when simulating rare events in random seaways. These events require extensive, long-run simulations across different loading conditions, wave environments, and operational scenarios, which makes computational efficiency a bottleneck.

This study investigates the use of massive parallelization on Graphics Processing Units (GPUs) to accelerate time-domain simulations for DSA. The approach is based on weakly nonlinear models, where radiation and diffraction forces are treated as linear, while restoring and Froude-Krylov forces are computed non-linearly based on the instantaneously submerged hull surface at each time step. We implement a 3 Degrees of Freedom (3DOF) simulation model, incorporating strip theory for radiation and diffraction, while leveraging Nvidia's Compute Unified Device Architecture (CUDA) platform to parallelize the computation of nonlinear restoring and Froude-Krylov forces.

The objective of this work is to showcase the implementation of GPU parallelization at a conceptual level, evaluate its computational performance, and propose areas for further optimization. The mathematical foundation builds on existing models but is reformulated for GPU parallelization to ensure faster, more efficient simulations for early-stage ship design and operational decision-making.

References

IMO MSC.1/Circ.1627. (2020). *Interim Guidelines on the Second Generation Intact Stability Criteria*.

Ports as Energy Hubs: Renewable Energy Integration and Distribution

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Abstract

Ports play a critical role in energy transition, serving as essential nodes in global trade. As they evolve into renewable energy hubs, it is crucial to integrate renewable energy systems (RES) and advanced energy distribution technologies. This study presents a review of the technical and infrastructural demands on ports as they evolve into renewable energy hubs. Emphasizing the need for smart energy management systems like microgrids and energy storage solutions highlights their essential role in balancing fluctuating renewable energy supplies. Furthermore, the deployment of power-to-gas (P2G) technologies, focusing on hydrogen production and distribution, is explored as a key element in transforming port energy systems. Emphasis is placed on the integration of offshore wind, solar, and tidal energy systems, alongside the deployment of advanced energy storage mechanisms to address volatility in supply. Multi-fuel infrastructure capable of supporting sea and landside transportation is examined, alongside renewable energy and carbon capture, utilization, and storage systems, which are pivotal in achieving decarbonization goals. This review provides a roadmap for policymakers and industry stakeholders to address the challenges and opportunities in the energy transition of ports.

Enabling Fuel Flexible Solid Oxide Fuel Cells (SOFCs) in Maritime Applications: 1D Simulation of SOFC System

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Abstract

As a promising green technology high-efficiency energy convertor, the interest on fuel cells have been remarkably increased. The Solid Oxide Fuel Cells (SOFCs) provides a wide range of advantages and features, especially its fuel flexibility. For SOFC's operation the Balance of Plant (BoP) has to be considered, designed as well optimized. In order to analysis such a complex system, the 1D numerical simulation plays a crucial role to examine the interaction between each component, the influences of changed parameters and the possible scenarios for the components' layout. This paper aims to create the modeling of BoP for SOFC in the 1D simulation program – Dymola. The early simulation results based on natural gas as fuel are validated regarding the selected literature. The proved configuration is implemented to investigate further the effects of input parameters at the system inlet such as pressure, temperature, fuel concentration as well fuel/air ratio. In addition, the specifications of components, for example, their sizes, reformer's operating temperature, compressor pressure ratio. are also studied. Finally, the performance of SOFC system is illustrated. The scenario of BoP's configuration is also presented. Moreover, the validated model will be further employed as a numerical tool for investigation of SOFC system fueling with the alternative fuels.

Quantifying Good Seamanship: A Framework for Risk-Aware Trajectory Planning in Autonomous Vessel Navigation

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Abstract

Autonomous vessel navigation plays a vital role in enhancing maritime safety and operational efficiency, particularly in congested and complex waterways. Navigators should always display “good seamanship”. This requires the ability to predict the future evolution of scenarios, assess risks, and plan several steps ahead. However, quantifying good seamanship remains challenging due to the limited ship maneuverability, multi-vessel encounters and near ground scenario. We have developed a framework for quantifying the good seamanship score based on the collision and grounding risk that takes into account the vessel kinodynamics constraint and the uncertainty of target vessel speed. This framework helps evaluate seafarers' actions in avoiding collisions and groundings. Utilizing a quantified measure, a risk-aware trajectory planning algorithm has been developed, which uses sampling-based methods to produce a safe and efficient trajectory, which is tested in various scenarios. Future efforts will address hydrodynamic effects such as wind and waves, consider target vessel future action, and explore advanced machine learning techniques to further improve the computation speed in scoring the good seamanship and generating the trajectory in complex scenario.

Research Questions:

- How can we define and quantify a reliable risk or good seamanship metric that accounts for all situations, including uncertainties, maneuverability limitations, different mission and vessel types, and hydrodynamic effects?
- Given these factors, how can we develop a real-time algorithm that generates the optimal trajectory for an autonomous vessel?

Addressing Seasickness of Technicians on board Working Vessels

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Abstract

In efforts to meet international goals for reducing greenhouse gas emissions while balancing rising energy demands, the number of installed offshore wind turbines has increased significantly over the past decades and is expected to continue increasing. As the number and size of offshore wind farms grow, so does the demand for technicians and crew transfer vessels (CTVs) for maintenance. Given the hazardous working conditions, the well-being and focus of technicians are essential for preventing catastrophic incidents. Despite being an age-old issue, seasickness remains a relevant challenge in the maritime industry that warrants further investigation. In addition to improving the work environment for technicians, assessing motion sickness on CTVs can also provide financial advantages for wind farm owners.

This project aims to develop a machine learning tool capable of estimating and predicting the probability of seasickness based on weather forecasts and sea conditions. The tool will be part of a decision support system for determining whether to sail. By accounting for the probability of seasickness, the decision-making process will be more considerate compared to the fixed parameters and rules of thumb currently used to assess the operational limits of CTVs.

The presentation at the DNV Nordic Maritime Universities Workshop will provide an overview of the project and the progress made so far. Key findings from a systematic literature review on existing methodologies for quantifying motion sickness will be highlighted, emphasizing the need for further research in this area. Additionally, data collected from motion sensors and a seasickness display installed on an operational CTV will be presented, along with analyses conducted to date and future plans for developing the predictive model.

Development of a mission-suitability assessment framework for marine vehicles (M-Score)

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Abstract

The traditional evaluation of marine vehicles' mission suitability has primarily centered around isolated functional requirements specific to the vehicle's primary purpose. For example, military and public vessels have historically been assessed based on attributes such as speed, range, hull strength, or deck space. These parameters, while vital, often overlook the broader, multifaceted nature of real-world operations.

Today in the maritime world, there is no systematic method or measure to answer the question “How fit is this marine vehicle for its intended purpose?”. Although many studies have been conducted to enhance the capabilities of marine vehicles, predominantly in the military sector [1], there are no established methods to quantify and rank their mission suitability or readiness levels.

This study aims to develop a comprehensive framework to assess the suitability of marine vehicles for various missions. The proposed framework introduces a scoring system that considers a wide range of complex, interrelated factors, related to vehicle characteristics, environmental conditions, human elements, societal impacts, and economic considerations. The framework will provide a systematic approach to identifying, classifying, and assessing the mission requirements of marine vehicles. The functional requirements for each mission requirement are also outlined based on the developed mission requirements. By considering the functional requirements and the vessel capabilities ,a mission suitability classification matrix is developed. Additionally, an integrated method for assessing functional requirements is introduced, allowing for the scoring and ranking of mission suitability for different types of vessels. The framework is empirically validated by applying it to various operational marine vehicles.

This framework will offer a versatile tool for various applications, including procuring, designing, and modifying marine vehicles. It also aids stakeholders like owners, designers, builders, and operators in decision-making. The framework's innovative scoring system enables the comparison of marine vehicles and assesses their suitability for specific missions. It represents a significant advancement in assessing marine vehicle fitness and mission readiness, improving decision-making for stakeholders.

The presentation will provide an overview of the study as an early stage of the PhD project and guide the audience through the key objectives.

References

[1] Aguiari M, Gaiotti M & Rizzo CM (2022). A design approach to reduce hull weight of naval ships, Ship Technology Research, 69:2, 89-104, DOI: 10.1080/09377255.2021.1947666.

Separating the impact of Hull and Propeller Degradation on Ship Resistance with Thrust Measurements

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Abstract

This presentation will go through the results and methods of a Master's thesis conducted in Spring 2024. The condition of the hull and propeller significantly influences a maritime vessel's fuel efficiency. Today, it is common practice to use advanced performance software to estimate performance changes caused by hull and propeller degradation, based on Noon reports and Autolog data. However, this approach approximates only the combined effect of hull and propeller degradation. Separating hull and propeller performance can improve decision-making for operators and stakeholders by providing better tools for planning dry docks, propeller polishing, or hull cleaning.

This thesis was conducted in cooperation with Vessel Performance Solutions ApS (VPS) to investigate the possibility of separately assessing a ship's hull and propeller condition while in service. Autolog data, including torque and thrust measurements, are used to achieve this. The method follows an approach suggested by Erik van Ballegooijen [1], with modifications to the speed-power baseline for the vessels. Two new baselines are proposed: the first is based on Vesper (VPS's in-house ship performance software), and the second is derived from propeller open-water curves and model tests. Results indicate that separation of hull and propeller effects can be achieved when sensor data is correctly calibrated and reported.

References

[1] Erik van Ballegooijen and Torben Helsloot. "An Approach to Monitor the Propeller Separately from the Hull". In: 4th HullPIC, Gubbio (2019).

Free surface wave problems via physics-informed neural networks (PINNs)

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Abstract

The Physics-Informed Neural Networks (PINNs) are introduced to solve the linear wave problem described by the potential theory. In the proposed PINNs framework, both soft and hard enforcing of boundary conditions (BC) at the bottom and sides of the wave domain are proposed. Two scenarios of solving the linear wave problem are investigated to find suitable PINNs architectures. Firstly, the free surface wave is considered to be completely defined, and the distribution of wave velocity components is used to evaluate the accuracy of PINNs using various combinations of BCs. Incorporation of the free surface and bottom BCs as soft constraints resulted in an average error of less than 4% in the calculation of velocity components. Incorporation of the Periodic BC (PBC) as a soft constraint reduced this average error to below 0.1%. The hard PBC is found to be slightly more computationally expensive. This study also explores the design of trial functions to implicitly impose the Kinematic Bottom BC (KBBC). While these trial functions strictly satisfied the KBBC, they limited the solution space, increasing the average error up to 10 times. Secondly, the wave angular frequency, ω, is considered as an unknown parameter, which is expected to be estimated by the PINNs. It was found that the proposed PINN framework can estimate the value of ω with an average error of 2.1%. The design of these hard constraints should be carried out with special attention not to increase the total error and computational cost.

Integration of the h-AES method into the design process of large structures for the fracture mechanical evaluation of accidental loads

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Abstract

The finite element method plays an important role in the design process of large structures such as ships. With its help, the impacts of different loads on the structure can be estimated in a time and resource efficient manner. In order to optimise computing times, the aim is always to achieve the coarsest possible meshing with sufficiently accurate results which often means that the entire structure is coarsely meshed while individual components or areas are meshed finer manually during preprocessing. This approach has proven its worth but reaches its limits when evaluating the impact of highly dynamic accidental loads, such as explosions or collisions. Those loads are characterised by the occurrence of high stresses and strains resulting from high forces and pressures in a very short time and can therefore cause failure of the structure. When evaluating the impact of accidental loads, it is reasonable not to carry out the local mesh refinement manually but to automate it. One method for automated mesh refinement is to use h-adaptivity. With this method, the solver first solves the mesh generated manually by the user. In the next step, an adaptivity module checks the calculated solution on the basis of error criteria. If the error criteria are not met, the mesh is refined locally by adding further elements and the solver solves the mesh again. This process is repeated until all elements fulfil the error criteria.
In addition to local mesh refinement, the method used to simulate crack propagation also plays a decisive role for the evaluation of the impact of accidental loads. For large structures that are discretised using shell elements, the element deletion method is often used to simulate crack propagation. However, this method has the drawback of losing mass and energy by deleting individual elements which means that energy conservation is no longer guaranteed. An alternative to the element deletion method is the element splitting method in which critical elements are split instead of deleted, thus preserving mass and energy.
A novel approach that combines the advantages of the h-adaptivity with those of the element splitting method is the h-adaptive element splitting (h-AES) method. It has been developed to simulate crack propagation with simultaneous conservation of mass and energy more precisely, and has already been successfully validated for static Mode I and II loads in the 2D case.
The aim of this research project is to extend the applicability of the h-AES method and to gradually integrate it into the design process of large structures in order to improve the fracture mechanical assessment of accidental loads.

A Discrete Element Method-based Mooring Dynamic Solver for Complex Soil–Cable Interaction

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Abstract

We present a novel implementation of a dynamic mooring solver accelerated by a dual-GPU platform, which can potentially host large systems for offshore energy, such as clusters of wind and wave farms, including shared mooring systems. Using the open-source Chrono DEM-Engine (Zhang, 2024), a new “force model” has been conceived and implemented to incorporate anchoring systems leveraging a lumped-mass approach. Lumped-mass methods can be very accurate in simulating extremely flexible elements, for which properties such as axial and bending stiffness, non-linear mechanical features, added mass, and wave loads, can be conveniently modeled. Chrono DEM-Engine is based on the Discrete Element Method (DEM) to model solid contacts, and this has been customized via a force model to define non-repulsive force (i.e., tension) between pairs of DEM elements to hold each line’s geometrical properties. Such high-fidelity framework configures a unique environmnet for studying cable–soil interaction for floating structures, providing great reliability for soil representation. It can comprise either complex shaped meshes, or granular soils.

For performance assessment, this implementation is benchmarked using a case that has met great consensus among the community for line solver validations (Chen & Hall, 2022). Figure 1 visualizes a four-point symmetric mooring system connected to a floating box, comprising catenary lines. The reference paper provides data from load cells at two anchoring points, respectively the starboard front and the back lines, for which validation will be provided.

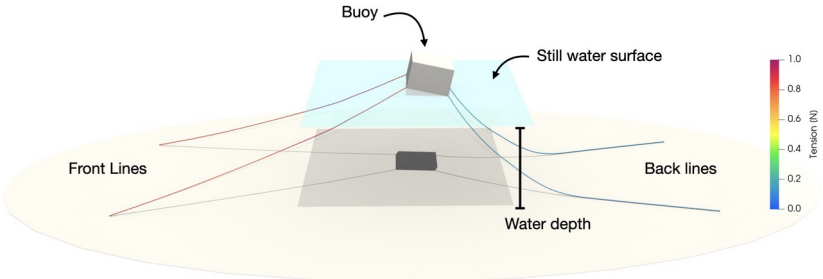


Figure 1 – Perspective view of a floating box with its 4-point symmetric catenary mooring system at its maximum tension in the front lines.

References

Zhang & Tagliafierro, et al. (2024). “Chrono DEM-Engine: A Discrete Element Method dual-GPU simulator with customizable contact forces and element shape.” In: Computer Physics Communications, p. 109196. DOI: https://doi.org/10.1016/j.cpc.2024.109196

Chen, H. & Hall, M. (2022). "CFD simulation of floating body motion with mooring dynamics: Coupling MoorDyn with OpenFOAM." Applied Ocean Research, 124, p. 103210. DOI: https://doi.org/10.1016/j.apor.2022.103210

Reduce fleet emissions through a bi-objective tramp ship routing and scheduling problem

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October 29, 2024

Abstract

We consider a bi-objective *Tramp Ship Routing and Scheduling Problem* (TSRSP) in which we plan routes for a fleet of tramp shipping vessels operating on a combined contract and spot market. The two objective functions are the maximization of profit and the minimization of GHG emissions. We present a unified model that can handle speed optimization, chartering costs, bunker planning, and hull cleaning. The single-objective model is solved by column generation, where the columns represent the possible routes of a vessel, while the master problem keeps track of the binding constraints. The pricing problem is solved efficiently using a time-space graph. Optimal integer solutions are guaranteed through Branch and Bound and Lagrangian Relaxations. The bi-objective model is solved using the ϵ-constraint method iteratively on the single-objective model.

The model's output is a curve called the Pareto front, describing the compromise between profit and emission. We compute the Pareto front for different historical scenarios, showing that emissions can be reduced by 10% with less than a 2% profit decrease and by 20% with less than a 6% profit decrease. Then, we compute the Pareto front with alternative fuels like LPG and LNG, obtaining emissions reduction of over 30% with less than a 5% profit decrease. Finally, we compare the results with a GHG levy.

The presented results are relative to four up to nine months plans. For plans with three or fewer months, the compromise advantages exponentially decrease. This means that for this GHG emission reduction measure, longer than usual and systematic planning are necessary to reduce emissions without impacting profits.

Keywords: Tramp shipping; Routing and scheduling; Column Generation; Bi-objective; Sustainability

Funding: Danish Maritime Fund, Grant 2022-058, and Ultrabulk

Investigating Material Behavior and Fatigue Strength of Welded Joints at Cryogenic Temperatures

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Abstract

The maritime industry is increasingly adopting alternative fuels, such as hydrogen, to support sustainable and environmentally friendly operations. This transition necessitates the increased use of cryogenic materials for the safe storage and transportation of these fuels at extremely low temperatures. Despite this growing demand, there is a notable lack of research on the mechanical behavior and fatigue strength of welded joints in such conditions. This research aims to address these knowledge gaps by investigating the mechanical properties of cryogenic materials and assessing the fatigue strength of welded joints exposed to cryogenic conditions.

To achieve these objectives, a comprehensive experimental study will be conducted to evaluate the effect of cryogenic conditions on the mechanical properties of cryogenic materials. By establishing how these properties change with temperature, the research will provide essential data to guide the selection of materials for safe and efficient cryogenic storage and transport. Additionally, since welded joints are critical components in maritime infrastructure, this research will emphasize the fatigue performance of welded joints exposed to cryogenic conditions by assessing the fatigue strength of welds at cryogenic temperatures.

The study will also explore the influence of various weld imperfections on the fatigue strength of welded joints in cryogenic environments, assessing how factors like defect size and location affect weld performance at low temperatures. Additionally, numerical models will be developed to predict the fatigue behavior of welds under cryogenic conditions. Advanced reverse engineering techniques will be used to create accurate 3D models of complex weld geometries, which will enhance the precision of fatigue behavior predictions.

The findings of this research will contribute to enhancing the safety, reliability, and efficiency of cryogenic storage and transportation systems in the maritime industry. By bridging existing knowledge gaps and providing practical insights, this study aims to facilitate informed decision-making in material selection and welding practices for cryogenic applications, supporting the adoption of alternative fuels and promoting sustainable operations within the maritime sector.

Potential for Hybrid Propulsion Systems for a selected fleet in Germany

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Abstract

The integration of inland and coastal shipping represents a crucial step towards the enhancement of supply chain sustainability. In comparison to other modes of freight transport, they offer the lowest emissions per transported unit of volume. Nevertheless, the majority of these vessels are currently still operated with diesel-powered propulsion systems. In order to further reduce the environmental footprint, a shift to alternative means of propulsion is necessary. The lower volumetric energy densities of alternative fuels, coupled with their scarcity, necessitate meticulous planning and adjustments to replenishment schedules. This places additional strain on operations when competing with more flexible transport options, such as road-based freight transport.

Therefore, this paper investigates the potential for hybrid propulsion systems in vessels without changes to their operational profile, with the objective of reducing the economic burden on the sector while contributing to more sustainable transport. AIS data is used in order to derive operational profiles of the considered fleet and to calculate their energy demand. From this, an estimation of the potential for alternative propulsion systems can be derived. Moreover, a linear optimisation model is employed to determine the necessary charger powers for a diesel-electric fleet.

The median achievable share of electric propulsion in relation to the total energy demand for a selected fleet of 2539 inland vessels running on diesel-electric propulsion systems falls between 12 and 32% depending on the assumptions made regarding energy storage and replenishment times. This equates to a potential for up to 0.7 million tonnes of CO_2 tailpipe emission savings. The necessary maximum charger power ranges between 100 and 200 kW, which is in line with the power requirements of state-of-the-art charging systems for road vehicles.

In conclusion, it can be demonstrated that hybrid alternative propulsion systems pose a great opportunity to enhance the sustainability of inland and coastal shipping, without necessitating alterations to the vessels' operational profiles. A network of replenishment stations (e.g. electric chargers) is however necessary in order to facilitate this. The requisite charging powers fall within the same range as those typically required for road vehicles.

Numerical and Experimental Investigation of Noise-generating Mechanisms for a Full-Scale Propeller

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Abstract

Underwater Radiated Noise (URN) from ship propellers is a topic of growing interest in the scientific community due to its adverse impact on marine life and due to upcoming maritime noise regulations. In the acoustic far-field, the sound radiation properties of the propeller are similar to that of an acoustic monopole, but it is often unrealistic to characterize the propeller as a constant-strength monopole due to the inherently different noise-generating mechanisms that constitute the radiated sound field from a propeller. Depending on the propeller design and operating condition, the relative importance of these noise-generating mechanisms can vary significantly. Hence, from the perspective of the propeller designer, it is important to understand the contribution of various noise-generating mechanisms to the overall radiated noise, particularly those related to propeller cavitation. The current project aims to address this design problem, through numerical predictions of URN and their validation through experimental measurements.

DoLPHiN is a numerical implementation of the solid-surface Ffowcs-Williams Hawkings (FWH) acoustic analogy, that includes reduced-order models for cavitation acoustics. While it can predict acoustic radiation from cavity surfaces, the accuracy depends on how well the hydrodynamic flow solver can predict cavitation inception and its extents. In the experimental campaign, we aim to identify the cavitation inception speed for different cavitation types through simultaneous visual observations, near-field pressure pulse measurements and far-field noise measurements. By choosing appropriate propeller loading conditions that excite different forms of cavitation, the obtained measurements are intended to serve as valuable data to calibrate the numerical flow model and to validate the numerical acoustic model. The validation exercise is expected to improve the reliability of the numerical URN prediction tool, thereby enhancing its utility for propeller design applications.

Parallel distributed collision avoidance with intention consensus based on ADMM

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Abstract

This research presents an approach to the problem of collaborative collision avoidance of autonomous inland ships. We propose a distributed model predictive control algorithm that allows ships to negotiate their intention to collaboratively avoid collisions directly. Furthermore, a new method is introduced to better shape ships' behavior in order to follow traffic regulations. The simulation results illustrate that the proposed algorithm could significantly increase ship safety navigation. Moreover, it also shows that the solution proposed by our algorithm complies with waterway traffic regulations except in complex situations when other safe solutions are negotiated.

Deep Reinforcement Learning for Revenue Management under Uncertainty in Master Stowage Planning on Container Vessels

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Abstract

Advanced planning policies obtained by machine learning have shown promising results in solving well-known combinatorial optimization problems in transportation and logistics. However, a significant challenge arises when dealing with complex action spaces in realistic planning, where it is less straightforward for machine learning models to generate feasible actions. A relevant and complex example is master stowage planning on container vessels, which plays a crucial role in global trade and the green transition. This planning problem aims to maximize cargo revenue and minimize operational costs while addressing strict constraints and demand uncertainty. To tackle this challenge, our paper introduces a deep reinforcement learning framework with a general feasibility layer to solve a novel Markov decision process of master stowage planning under demand uncertainty. The experimental evaluation shows that our architecture efficiently finds feasible solutions for a multi-stage stochastic optimization problem, which is intractable using traditional benchmark methods from combinatorial optimization. Our approach demonstrates the potential of advanced planning policies to tackle complex, real-world problems, with implications for global trade and sustainability.

Short Sea Shipping Route Segmentation and Bayesian Optimisation of Engine Power Allocation

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Abstract

To enhance energy efficiency, ship operators and regulators are focusing on meeting regulations, cutting costs, and reducing emissions, affecting all phases of a voyage, from planning to real-time sailing adjustments. Although speed optimisation techniques are widely studied, they often prove impractical because ship propulsion is controlled by engine parameters. Additionally, ships frequently follow similar routes between consistent ports, making this an effective input for the optimisation framework. This approach narrows the optimisation goal to finding the best engine power allocation for minimising fuel consumption. A two-stage power allocation optimisation is introduced for short sea voyages. The first step divides the route into segments based on a metocean data ensemble from a reference feasible voyage. The PELT algorithm detects changepoints, clusters sea states, and segments the route. In the second stage, Bayesian Optimisation (BO) identifies the optimal power allocation for each segment. The initial suboptimal power allocations for the BO are set using Dynamic Programming, ensuring the ship reaches its destination within the estimated time of arrival (ETA) while improving fuel efficiency. The approach was validated through numerical experiments on voyages across the Baltic Sea, North Sea, and English Channel, with the optimal power allocation resulting in fuel savings of 5% to 10% with less than an hour's deviation in ETA when compared to real data measurements.

Keywords: Voyage Optimisation, Route Segmentation, PELT, Dynamic Programming, Energy Efficiency, Power Allocation Optimisation, Bayesian Optimisation

A Digital Twin Framework for Massively Parallel Simulations of Water Waves interacting with Large-Scale Offshore Structures

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Numerical computations of wave-wave and wave-structure interactions can be extremely costly and time-consuming when the simulations are large-scale (large domains and/or multiple complex bodies) and over long periods of time. Hence, it stresses the importance of numerical strategies that are efficient and scalable. This work focuses on implementing a massively parallel spectral element solver based on fully nonlinear potential flow models [1]. This solver constitutes the basis for a new Digital Twin simulation framework that will enable the simulation of offshore structures in a virtual environment. The framework aims to reproduce realistic sea state conditions and include offshore structures, such as floating wave energy devices and wind turbine structures. The long-term vision is to be able to support methods that allow the integration of sensor data in the modeling. The numerical backbone of the model is currently developed using the open-source high-order finite element framework, Firedrake [2], which supports the implementation of state-of-the-art finite element methods. The Firedrake software is designed to ease the steps from continuous governing equations to discretized systems and simulations. The spectral element method has been studied for wave and wave-structure simulations for over a decade at DTU Compute [1,3-4]. Ultimately, the proposed framework aims to resolve wave-related problems in full spatial dimensions (3D) of various levels of fidelity with emphasis on incompressible free surface flow modeling and floating offshore structures. The presentation will motivate the project's first results and showcase relevant advances in the ongoing implementations. The work is supported by COWIFonden (Grant A-165.19).

References:

[1] Engsig-Karup, A. P., Eskilsson, C., & Bigoni, D. (2016). A stabilised nodal spectral element method for fully nonlinear water waves. Journal of Computational Physics, 318, 1-21.

[2] Rathgeber, F., Ham, D. A., Mitchell, L., Lange, M., Luporini, F., McRae, A. T., ... & Kelly, P. H. (2016). Firedrake: automating the finite element method by composing abstractions. ACM Transactions on Mathematical Software (TOMS), 43(3), 1-27.

[3] Visbech, J., Engsig-Karup, A. P., & Bingham, H. B. (2024). Solving the complete pseudo-impulsive radiation and diffraction problem using a spectral element method. Computer Methods in Applied Mechanics and Engineering, 423, 116871.

[4] Engsig-Karup, A. P., Monteserin, C., & Eskilsson, C. (2019). A mixed eulerian–lagrangian spectral element method for nonlinear wave interaction with fixed structures. Water Waves, 1(2), 315-342.

An Experimental Study on the Hydrodynamic Drag of Flexible Seaweed Mimics in a Uniform Current

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Abstract

An extensive series of model tests was conducted to investigate the hydrodynamic drag of flexible seaweed mimics in a uniform current. The material selected for modelling kelp blades was silicone rubber (E=5.36±0.4MPa, ρ=1264.5±66.0kg/m^3). The primary non-dimensional parameter examined is the Cauchy number (Ca), which represents the ratio of external drag to the internal stiffness of the blade. The experiments spanned a wide range of Ca values, from 10^2 to 10^6, revealing two distinct motion regimes: the static regime and the fluttering regime.

In the static regime, a balance between drag force and bending stiffness (buoyancy as well for non-neutrally buoyant structures) leads to a static blade position and a reduction in drag. Upon reaching a critical Ca value, fluttering occurs, and the blade motion becomes dynamic. At this point, a noticeable change in drag reduction is observed, shifting from a monotonically decreasing trend to an almost constant value.

The study found that drag reduction and critical Ca depend little on the blade's aspect ratio. However, the buoyancy parameter (the ratio of buoyancy to bending stiffness) plays a significant role in both regimes. A smaller buoyancy parameter results in a lower critical Ca and greater drag reduction in the static regime, but less drag reduction in the post-critical regime. Additionally, the line density of the blades (number of blades per unit length) significantly affects both the critical Ca and the drag reduction after the critical point. Based on the experimental results, a rough estimate of the drag coefficient C_D is also provided.

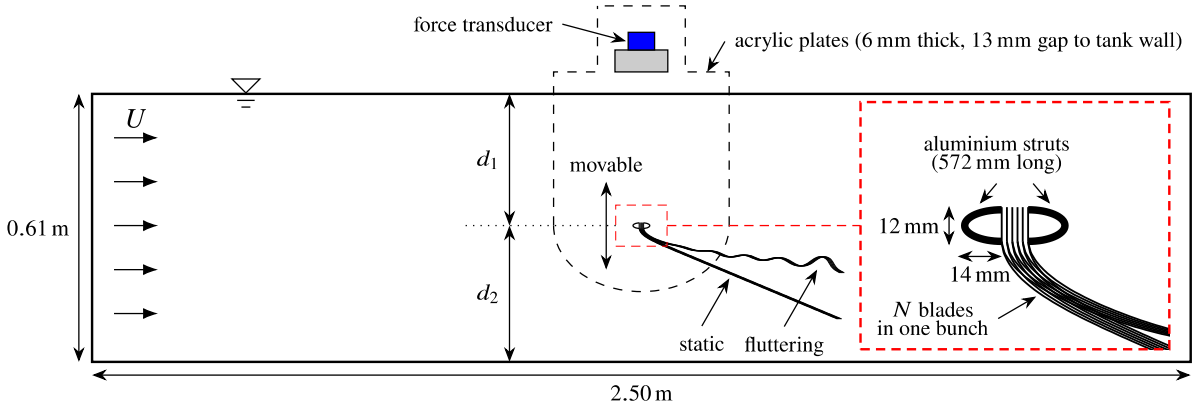


Figure: sketch of the test section and setup in the Circulation Water Tunnel (CWT) at NTNU.

Further steps towards generalisation of green water probability distributions on ships with forward speed

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Abstract

Green water is wave-structure interaction phenomenon that can lead to ship damage, or worse. Fundamental contributions to the phenomenology of green water for structures without forward speed were made by Buchner (2002) and Greco et al. (2005). Guedes Soares and Pascoal (2005) considered probability distributions of the occurrence of water on deck among other parameters, also for zero mean forward speed.

Forward speed was specifically included in the experiments conducted by Boon and Wellens (2022b). The setup of the experiments – with the ship supported at a mean position in the middle of a wide tank with free surface waves superimposed on a mean current – was such that single tests with multi-hour measurement times were possible. The data-set resulting from the measurement campaign has been shared *open access* as part of Boon and Wellens (2022a), signals of which were analyzed in order to find probability distributions of relevant parameters. Whereas probability distributions are typically fit to higher-probability data and extrapolated to find parameter values at lower probabilities, it was found by fitting various extreme value distributions to also the high values at lower probabilities that the time between green water occurrences follows an exponential distribution and that the maximum green water impact pressure follows a Fréchet distribution.

The research question that we would like to address in the discussion is what the effect of forward speed is on the value of the green water impact pressure. Because Computational Fluid Dynamics is highly suitable for changing a single parameter while keeping other parameters the same, we have taken simulations of a numerical method that is based on the Navier-Stokes equations, using an improved Volume-of-Fluid (iVOF) approach for transporting the free surface, and have validated them against tests in the aforementioned data. Additional simulations were run, in which only the forward speed of the ship was changed. Results of these simulations will be discussed at the workshop.

References

Buchner, B., 2002. Green Water on Ship-Type Offshore Structures (Ph.D. thesis). Technische Universiteit Delft, URL https://repository.tudelft.nl/islandora/object/ uuid%3Af0c0bd67- d52a- 4b79- 8451- 1279629a5b80.

Greco M, Faltinsen, O.M. and Landrini M., 2005. Shipping of water on a two-dimensional structure. *Journal of Fluid Mechanics*. 2005;525:309-332. doi:10.1017/S0022112004002691.

Guedes Soares, C., Pascoal, R., 2005. Experimental study of the probability distributions of green water on the bow of floating production platforms. J. Offshore Mech. Arct. Eng. 127 (3), 234–242. http://dx.doi.org/10.1115/1.1951773.

Boon, Anna, Wellens, Peter, 2022a. Data from Long-Running Green Water Experi- ments in Various Sea States At Model Scale. 4TU.ResearchData, http://dx.doi.org/ 10.4121/21031981.v1, URL https://data.4tu.nl/articles/dataset/Data_from_long-running_green_water_experiments_in_various_sea_states_at_model_scale/21031981.

Boon, A.D., Wellens, P.R., 2022b. Probability and distribution of green water events and pressures. Ocean Eng. http://dx.doi.org/10.1016/j.oceaneng.2022.112429.

Early-stage PhD presentation: Wind as primary ship propulsion

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Abstract

The research is grounded in the need to cut rapidly and drastically environmental impacts from shipping and investigates the potential of primary wind propulsion as a credible technology to harvest the abundant, renewable and free energy from the wind, with a low well-to-wake footprint.

Wind propulsion devices are increasingly available on the market, and there is a widening research interest in the aero-hydrodynamic evaluation of their performance and design. The matter is most often focused on wind as auxiliary propulsion, at single ship scale. The objective here is to tackle the issue the other way around by evaluating feasibility of sails as primary propulsion from a systemic fleet or global perspective. The ambition is to contribute to bridging the gap between initiatives to use wind as auxiliary propulsion in conventional trade, and existing alternative, small-scale, primarily wind-driven trade systems, by investigating:

1. Suitable strategies to establish a mainly sail propelled fleet, including design, operational and commercial aspects
2. Conditions for viability, uptake and diffusion of such a sailing fleet in terms of trade patterns, business, regulatory and geographical settings
3. Impacts of future policy alternatives on the global uptake of wind propulsion

The methods for exploring the aforementioned challenges will be presented and discussed.

The Atmospheric Boundary Layer and its modelling in CFD for Ship Wind Propulsion Applications

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Abstract

In the renewed interest to harvest wind as an energy source to propel cargo ships, knowledge of the nature of the wind at sea is an important aspect for a reliable development of Wind Propulsion Technologies (WPTs). In general, a flow moving over a surface will generate a thin layer called boundary layer, at the surface, where viscous forces are not negligible. In the case of the atmosphere, the Earth surface can generate a boundary layer with depths between 10 to 3000 meters. This boundary layer is fully turbulent, causing a rapid turbulent transport and mixing, it has anisotropic turbulence, and strong friction effects are present on the surface. The depth, wind velocity profile, and turbulent characteristics of the Atmospheric Boundary Layer (ABL) are driven by the thermal stratification within this boundary. The ABL is typically categorized in three main stratifications: nominally unstable, neutral and stable.

Neutral stratification is mostly used in engineering applications; this stratification has an adiabatic temperature gradient and does not need a buoyancy term. This description can be found in Computational Fluid Dynamics (CFD) simulations about: offshore floating facilities and their wind loads, urban areas for pedestrian comfort and loadings, and offshore wind farms for performance prediction and interaction effects of wakes.

The neutral stratification is used because of its modelling simplicity but does not represent the entire spectrum of wind conditions. Hence, since all the stratifications are observable throughout a year on open water, and have a different energy content and turbulence characteristics, it is important to consider them for reliable performance predictions and load calculations for ship wind propulsion applications.

A series of CFD simulations of a two-dimensional empty domain are setup with an atmospheric boundary layer, represented by a power law, prescribed at the inlet together with the semi-empirical vertical profiles for the turbulent kinetic energy k and the specific dissipation rate of turbulence ω, as suggested by the Architectural Institute of Japan. The three different stratifications are obtained by adopting three different values for the power law exponent describing the ABL. The results show how the velocity profiles are well preserved throughout the domain, while the turbulence quantities are either under- or over-estimated.

Impact of attention and working memory on maritime navigational performance and situational awareness

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Abstract

This study explores the potential impact of attention and working memory assessments e.g., N-back test on navigational performance and situational awareness in maritime roles. While cognitive tests are widely used in the selection of highly attentive professions such as airline pilots, they are not commonly integrated into maritime navigation. Given the critical importance of attention and situation awareness in safe navigation, it is essential to explore whether these cognitive abilities have an influence and can be used as predictors of performance at sea.

Through a series of experiments, this study examines whether individuals who score higher on attention tests demonstrate better navigational performance and greater situational awareness compared to those with lower scores. Participants with varying levels of attention and working memory capacity will complete a series of navigation tasks under the definded navigational sceniaros while being assessed for their awareness of the environment and decision-making accuracy. The presentation presents the test procedures and aims to establish whether cognitive assessments could have value in selecting and training maritime navigators.

How can we enhance maritime simulation training using insights from comprehensive accident data?

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Abstract

On 26 March 2024, the container ship *Dali* collided with a support pillar of the Francis Scott Key Bridge, which caused a significant portion of the bridge to collapse, with one of the spans falling directly onto the ship's forecastle. The accident tragically resulted in the deaths of six construction workers and left one crew member injured, while the overall damages led to massive economic losses. Navigating in bridge environments is inherently challenging due to several factors. The narrow navigable waters require precise maneuvering to avoid collisions with both the bridge structure and other vessels. In addition, communication with other ships, the effects of currents, and the influence of environmental factors like tides further complicate operations. The situation becomes even more hazardous in poor weather conditions, such as strong winds, heavy rain, or low visibility, which impair judgment and increase the risk of accidents.

Given the challenges faced in bridge environments, specialized training for seafarers is crucial. To improve maritime simulation training and identify the key skills required for bridge operations, we propose a data-driven approach based on accident analysis. We first examined data from 54 ship-bridge collisions to determine their root causes. These insights were then aligned with the skills outlined in the STCW Convention, helping us identify the most critical skills. These findings offer instructors scientific guidance in designing training scenarios, enhancing both the effectiveness and precision of the training.

Integrated Shipbuilding Framework for Decarbonisation Based on Digital Twins

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Abstract

Given the potentially increasing global shipping emission share (from 3% to 13%), the whole shipping industry faces an urgent challenge to mitigate the carbon emissions within the entire ship lifecycle. The Maritime Forecast 2050 report from DNV pinpoints Digitalisation and Decarbonisation as the most transformative forces that will shape the future of the shipping industry. Digitalisation, through cutting edge technologies like Digital Twins (DTs), has the potential to revolutionise ship design and production, thus driving decarbonisation. In the current shipbuilding process, design plans are passed directly to production, with most designers focusing solely on decarbonisation through propulsion system and hull shape design, overlooking the production phase entirely. Consequently, ship manufacturers are unable to offer any feedback during the design phase, creating a significant decarbonisation gap in the shipbuilding process. It is highlighted that DTs possess the potential to seamlessly integrate ship design and production starting from the early design phase. This integration is facilitated through a shared set of comprehensive DT models, leveraging the holistic and high-fidelity capabilities inherent to DT technology. However, extensive literature reviews indicate that the application of DTs in shipbuilding is still in its infancy, particularly in the integration of ship design and production. This research aims to address the challenge by developing a DT-based integrated shipbuilding framework for decarbonisation. A concise literature review of the state-of-the-art DT applications in integrating ship design and production will be provided. Following this, the proposed DT-based framework will be then presented, detailing its specific modules and data transmission pathways. Lastly, our future plan for conducting a pilot case study to validate the framework and the related DT architecture basis will also be shared.

Research Questions:
What is the current status of DT applications enabling decarbonisation in shipbuilding?
How can DTs integrate ship design and ship production for decarbonisation?

Keywords: Ship Design; Ship Production; Digital Twins; Decarbonisation; Shipbuilding

FUTURE SHIP DESIGN FOR DECARBONIZATION

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The maritime industry faces increasing pressure to reduce its greenhouse gas (GHG) emissions, particularly CO_2, which accounts for 2.9% of global emissions. Without action, shipping emissions could rise by 90% to 130% by 2050. In response, the International Maritime Organization (IMO) introduced a revised GHG strategy, aiming for net-zero emissions by 2050. The DNV *Maritime Forecast to 2050* highlights digitalization and decarbonization as key drivers expected to significantly transform the shipping industry. Wind-Assisted Propulsion Systems (WAPS) offer a promising decarbonization solution by harnessing wind—an inexhaustible, zero-emission energy source—while digitalization, utilizing tools like Digital Twins (DTs), can further enhance ship design. However, significant gaps currently exist in the adoption and integration of WAPS and DT technology within the maritime industry. Limited studies have explored WAPS's potential for decarbonization, and technical risks have not been sufficiently addressed, contributing to shipowners' hesitation. Meanwhile, the application of DT in ship design remains under-researched, with a lack of in-depth modelling approaches and insufficient demonstration of the virtual-to-physical connection between digital twins and real-world systems. Consequently, this thesis aims to address these gaps by focusing on these key research questions:

1. “How do WAPS contribute to decarbonization goals, and what are the challenges associated with their implementation in the maritime industry?”
2. “What role can DTs play in supporting the development of sustainable ship designs, particularly for WAPS?”

To answer these questions, this research will integrate DT technology into ship design, with a focus on WAPS. A DT model will be developed as a testbed for optimizing future ship design and performance. Using a 7-meter model as a case study, experimental data will be collected to test, refine, and validate the DT models. Insights gained will provide guidance on the role of both DT technology and WAPS in achieving decarbonization, helping stakeholders make informed decisions about adopting these innovative technologies. Ultimately, these findings will support the IMO’s emissions reduction targets by promoting sustainable practices in ship design and operations.

Hydrologic Riverbed Generation and Operational Analysis for Autonomous Vessels in Meandering Inland Waterways

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Abstract

Confined water is a critical factor affecting the performance of autonomous vessel navigation. While existing research primarily focuses on improving algorithms for tasks such as collision avoidance or track piloting, the influence of environmental effects, particularly confined waterway itself, is often neglected or overly simplified. This study presents a novel method for the rapid generation of arbitrary inland waterway scenarios, which is crucial for validating the performance of autonomous vessels under dynamic confined water conditions. A new formula is proposed to model river hydraulics, such as cross-sections shifting and current fields for meandering waterways, incorporating the effects of deposition and erosion on bathymetry. The position of maximum depth has been validated through field measurements from a natural river in the United States. The formula successfully captures the general trend of lateral migration of the thalweg between inner and outer bends. Furthermore, a ship operational analysis was conducted to examine the impact of arbitrary waterway configurations on vessel maneuvering performance. The results indicate the significant influence of hydrodynamic effects on vessel steering behavior, emphasising the importance of incorporating bathymetric and current data into control system design to ensure navigational safety for autonomous vessels in confined waters.

Identification of tank cleaning operations at sea based on machine learning method using AIS data

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Abstract

The increase in transportation of liquid bulk chemical substances, such as biofuels, has led to a rise in chemical pollution incidents in Swedish waters and the Baltic Sea. Illegal discharges of wash water and chemicals from operational tank cleaning of tankers exert significant pressure on the marine environment. However, regulating tank cleaning is challenging, and the behavioral patterns of illegal discharges are difficult to define. Current regulations, mainly within the IMO MARPOL Annex II, related to discharge of tank cleaning residues are intricate and leave room for different interpretations. One possible approach to identify potential tank cleaning activities is by analyzing Automatic Identification System (AIS) data and the spatiotemporal information it contains. This study proposes two methods to investigate abnormal behaviors of ships: a) examining possible causes of abnormal AIS signal missing (e.g., intentional shutdown of AIS devices or technical malfunctions), and b) applying the machine learning algorithm to detect abnormal trajectories (such as unnecessary sharp turns) to identify potential tank cleaning operations. By analyzing observed illegal discharges data, the characteristics of ships at the time of spills, such as trajectory patterns, speed, and loading conditions. Additionally, we cluster AIS data from ships granted prewash exemptions to establish normal sailing characteristics (intervals of AIS signal missing). Results show that missing AIS data messages are more frequent for the investigated chemical tankers compared to cargo ships. Even though it is not possible to judge how many of those AIS signals missing scenarios are related to tank cleaning, serious AIS signal loss, i.e., more than 30-50 minutes data while the ship is sailing faster than 5 knots, may be interpreted as suspicion of tank cleaning by looking at their trajectories. For chemical tankers granted prewash exemption, much fewer AIS missing data were observed (no serious AIS signal missing was observed). The implemented data mining methods are expected to support maritime authorities.

AI-Driven Ship Brain Framework for Intelligent Decision-Making in Sustainable Maritime Navigation

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Abstract

The concept of a Ship Brain introduces a transformative approach to enhancing maritime safety and efficiency through the integration of cutting-edge artificial intelligence (AI) technologies. By leveraging models for critical scenarios detection, ship motion prediction, ship residence prediction, and ship damage evaluation, this advanced system provides a comprehensive framework for intelligent navigation monitoring and decision-making support. The Ship Brain architecture integrates marine hydrodynamics, ship operations optimization, and emerging AI technologies to ensure safer and more sustainable maritime operations.

The primary functions of the AI-driven Ship Brain include predictive modeling for ship trajectories and motion under varying hydrometeorological conditions. This capability enables real-time testing and selection of safe maneuvering commands, such as adjusting speed and rudder angle, to prevent accidents like collisions and groundings. Additionally, the Ship Brain predicts fuel consumption and optimizes ship operations by identifying the most energy-efficient maneuvering strategies, contributing to significant cost savings and environmental benefits.

The system incorporates a multi-model framework where critical scenarios detection assesses navigational risks based on traffic distribution and environmental hazards. The ship motion prediction model provides insights into the vessel's spatial and temporal behavior, informing strategic operational adjustments. Furthermore, the maneuvering model evaluates ship dynamics, testing various control commands to ensure optimal performance. The safety evaluation component quantifies potential damage and ensures decisions prioritize crew and environmental protection.

In summary, the AI-driven Ship Brain exemplifies a holistic approach to intelligent maritime decision-making, combining domain-specific knowledge of hydrodynamics and operational optimization with AI advancements. By facilitating predictive, adaptive, and data-driven navigation, this system addresses the challenges of modern maritime transportation, fostering enhanced safety, operational efficiency, and environmental sustainability.

High-Fidelity Numerical Analysis of Hull-Wingsail Interaction

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Abstract

Wingsails represent an innovative approach to wind-assisted propulsion systems (WAPS) in maritime transportation, which have gained attention for their potential to significantly reduce fuel consumption and greenhouse gas emissions by harnessing wind energy more efficiently than traditional sails (Malmek *et al.*, 2024; Zhu *et al.*, 2023). However, the interaction between wingsails and the ship's hull remains an underexplored area, especially regarding the influence on propulsive performance, ship stability, and seakeeping ability. The unique aerodynamic and hydrodynamic characteristics of crescent-shaped wingsails, which provides greater thrusts, introduce complexities in how the generated lift and drag forces interact with the hull.

This study investigates the hull-wingsail interaction through simulations of a single wingsail, a bare hull, and a combined hull-wingsail system, focusing on a side wind condition with an apparent wind angle of 90 degrees. High-fidelity computational fluid dynamics (CFD) simulations based on improved delayed detached eddy simulation (IDDES) (Zhu *et al.*, 2024) are used to predict the wind loads on the wingsail and the hull-wingsail interaction.

Preliminary results reveal significant differences in the aerodynamic performance of wingsails when coupled with a hull, highlighting that the hull's presence substantially alters the flow field and thus the wingsail's efficiency. The hull creates strong horse-shoe vortices and extrudes the upstream flow, leading to higher streamwise flow velocities around the lower sections of the wingsail. This results in increased lift force coefficients. Additionally, the interaction between the hull and the wingsail modifies the pressure distribution around both elements, further influencing the overall aerodynamic forces acting on the system. These changes suggest that the wingsail's thrust and efficiency are strongly affected when its interaction with the hull is considered. Therefore, neglecting the hull-wingsail interaction might lead to an underestimation of propulsive performance.

Malmek, K., Larsson, L., Werner, S., Ringsberg, J. W., Bensow, R., & Finnsgård, C. (2024). Rapid aerodynamic method for predicting the performance of interacting wing sails. *Ocean Engineering*, *293*, 116596. https://doi.org/https://doi.org/10.1016/j.oceaneng.2023.116596

Zhu, H., Yao, H.-D., & Ringsberg, J. W. (2024). Unsteady RANS and IDDES studies on a telescopic crescent-shaped wingsail. *Ships and Offshore Structures*, 1-14. https://doi.org/10.1080/17445302.2023.2256601

Zhu, H., Yao, H.-D., Thies, F., Ringsberg, J. W., & Ramne, B. (2023). Propulsive performance of a rigid wingsail with crescent-shaped profiles. *Ocean Engineering*, *285*, 115349. https://doi.org/10.1016/j.oceaneng.2023.115349

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